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Survey

Aeronautical communication transition from analog to digital data: A network security survey

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ABSTRACT

The objective of this paper is to present a comprehensive survey of security challenges in aeronautical data communication networks. The civil aviation industry is currently going through an evolution of the air traffic management system. The aviation communication technologies are progressively shifting towards the use of digital data instead of analog voice for traffic control, airline business, and passenger onboard entertainment systems. This paper illustrates the *cause-to-effect* chain link starting from the modernization of the aeronautical communication systems and leading to the network security concern in the civil aviation. The general threats to air-ground communication are depicted and then categorized. The paper gives an overview of the civil aviation industry efforts for securing the future aeronautical data communications. The security mechanisms and protocols proposed for this purpose are discussed. Open research issues and challenges that have to be addressed in the security of current and future aeronautical data communication networks are presented in detail. The paper concludes with some improvement directions which can help to address those security issues. This survey can be used as a reference guide to first understand the factors that urge both the research community and the aviation industry to be concerned about network security in future aeronautical data communications. Also, it can be used as a first reading to have a global overview of network security issues, challenges and potential solutions in air-ground communication networks.

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1. Introduction

Aeronautical communications are about to shift the paradigm of digital data in near future. Thanks to IT (Information Technology) progress made in last decades, aviation industry

stakeholders are enhancing and expanding their networks to overcome problems related to voice radio communications, but also to modernize the ATM (Air Traffic Management) environment. In 1978, a digital data link system was introduced, namely ACARS (Aircraft Communication Addressing and Re-

porting System) [1], to essentially support radio voice systems, which were nearly running at their maximum capacity.

The term “data link”¹ is commonly used among the civil aviation community to represent digital communications between an aircraft and a ground station (i.e. an air traffic tower control, an airline operational control center). In 1983 ICAO (International Civil Aviation Organization) established the FANS (Future Air Navigation System) special committee, charged with developing operational concepts for the future ATM, including data link communications [2]. In 2001, EUROCONTROL (European Organization for the Safety of Air Navigation) launched the “LINK 2000+” program which is a set of implementation rules for data link networks and technologies in the European airspace [3].

The open and wireless environment of data link networks makes it vulnerable to serious attacks. Data link networks raise several security concerns for all users including pilots, air traffic controllers, airline staff, and even passengers. Thus, data link networks have to be protected from attacks which might have serious consequences in a critical infrastructure such as the aviation transportation system.

This paper presents a comprehensive survey of network security issues and challenges in future data link networks. Potential attacks and provided solutions are discussed. The paper explains the motivation and will of the aviation industry to modernize the aeronautical communications, which leads to network security concerns. In addition, the paper provides solutions to address some open security research issues and challenges in data link networks.

The rest of the paper is organized as follows. Section 2 provides a step-by-step description of ATM and CNS² (Communication Navigation and Surveillance) paradigms evolution towards digital data. This is helpful to understand why the industry is now putting so much efforts to provide secure communication solutions and mitigate related risks. Section 3 briefly shows how such a shift towards data link in air transportation systems can create network security issues. Section 4 summarizes the basic security and cryptographic primitives used later in the following sections. Section 5 provides an intruder model in data link environments and a classification of potential attacks on the aeronautical communication network. Section 6 presents an overview of the industry efforts regarding network security in data links and a state of the art of the provided security solutions. Section 7 discusses the main security challenges in data link networks. Finally, Section 8 provides three improvement directions to mitigate the identified security challenges in future aeronautical communication networks.

2. Evolution of aeronautical communication networks

2.1. Growth in air traffic load

Currently, the pilot and the controller mainly communicate via HF (High Frequency) or VHF (Very High Frequency) analog

voice radios, respectively in oceanic and continental areas [4]. When an analog voice radio communication technology is used, all pilots in the same sector and communicating with an air traffic controller are tuned to the same frequency. This can be challenging considering the expected air traffic growth. Statistical data on air traffic increase are very important in identifying trends within the air transportation industry. Long term forecast studies provided by Boeing predict a 5% growth rate of the world air traffic load between 2011 and 2030 [5]. This growth is due to many factors such as an increased aircraft manufacturers market, more competitive low-cost airlines, an increased passenger demand and the greater need for companies to provide a better service to their customers. Nowadays, the air traffic load is still increasing, leading to a congestion of the worldwide analog voice frequency allocated to the civil aviation.

2.2. Congestion of the aviation radio frequency

The civil aviation airband spectrum, which uses radio frequencies between 108 and 137 MHz (Mega Hertz), is already congested. A frequency saturation may likely delay the communication between the pilot and the controller and make them unreachable for a certain period of time. ICAO has shown its concern and expressed the need for a larger spectrum for future aeronautical air-ground communications systems [6].

As consequences could be catastrophic in such circumstances, many solutions have been provided in the past in order to address this frequency congestion issue such as optimizing frequency re-use, using a larger spectrum (since 1979, it has been extended from 136 to 137 MHz), or even splitting the radio spectrum into narrower bandwidths (50 kHz to 25 kHz channels). In 1994, it has been decided to reduce the channel spacing from 25 to 8.33 kHz (Kilo Hertz) under the ongoing the EUROCONTROL “8.33 kHz program” initiative [7].

However, dividing saturated air traffic sectors into smaller sectors implies few shortcomings. First, the number of available voice communication channels is still limited: there may not be enough frequencies when reaching a saturation point in a high density airspace in few years. Also, sector division requires coordination between air traffic controllers responsible for two neighboring sectors when an aircraft goes from a sector to another. Then, the pilot has to switch to the new channel frequency, test it, and turn-off the previous channel frequency.

Hence, air traffic sector division cannot be considered as an efficient long-term solution for radio voice frequency saturation. In order to address the air traffic growth and subsequent frequency congestion issues, the industry is progressively leaving analog voice at the expense of digital data communications.

2.3. Modernization of aviation communication technologies using data link systems

Data link systems seem to be an efficient conflict resolution strategy to:

¹ not to be confound with the OSI protocol layer 2 definition. In the rest of the paper, the term “data link” will be used as a reference to aeronautical digital communications.

² Navigation and surveillance are out of scope of this paper.

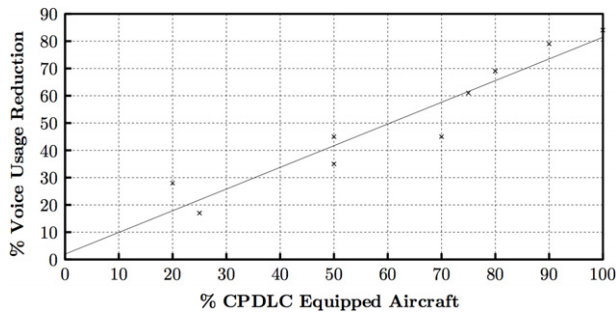


Fig. 1 – Voice usage reduction as a function CPDLC equipped aircraft.

- Reduce communication delay and increase voice channel availability. Indeed, data link systems will be used as complementary solutions to legacy and existing systems. As more communication systems are provided and used alternatively, the analog communication systems will be more available to usage. Besides, pilots are no longer required to wait until the carrier is free to talk.
- Enhance working environment for both pilots and air traffic controllers. For instance, aircraft will be equipped with DCDU (Data Link Control and Display Unit), which is an interface for sending and receiving text messages. Such a system provides a visual comfort to the pilot and a eventually a support to confirm the instructions they received using other systems.
- Limit misunderstanding and misinterpretation due to poor voice quality. Due to frequency congestion issues such as communication cutout or interferences, the pilot could understand wrong the instructions sent by the air traffic controllers or not receive them at all, which will likely lead to catastrophic consequences. As data link systems will limit the usage of legacy and analog voice communications, it might help in avoiding such aftermaths.
- Avoid signal corruption as, unlike analog voice transmissions, digital data solutions are not subject to the frequency congestion issues explained in Section 2.2.

Many aircrafts are already equipped for data-based communications with data link systems such as the CPDLC (Controller to Pilot Data Link Communication) system [8], which is an ATN (Aeronautical Telecommunication Network) [9] data link application that allows text-based message exchange between airline/air traffic ground facilities and the aircraft. Promising statistics showed that analog voice usage for operational services decreased in aircraft equipped for data-based ATS (Air Traffic Service) and AOC (Airline Operational Communication) applications. Shingledecker et al. [10] demonstrated that, for 100% CPDLC aircraft equipped, the radio voice usage decreased up to 84% as shown in Fig. 1. This kind of result points out the general trend in aeronautical communications: less voice and more data in the future.

2.4. Future aircraft data communication services

Future air communication services and their supporting systems will be promptly based on data link technologies to provide both operational services and non-operational services. According to the ICAO Annex 10 [11], operational services are categorized as follows:

- **Air Traffic Services:** ATS services support ATC (Air Traffic Control) messages between the pilot and the traffic controller. These services are provided by ATC ground entities performing specific ATS functions such as flight information, flight instructions, or departure/landing clearances. For instance, the DCL (Departure Clearance) service gives the permission to the pilot to take-off. The DYNNAV (Dynamic Route Availability) is another ATS service used by an ATC ground entity to indicate alternative routes to the crew during the flight [12];
- **Airline Operational Services:** Besides cockpit ATS services, AOC services are required for efficient CNS and ATM operations. This service category supports operational voice and/or data messages between the aircraft and the airline or airport operational staff. Usually, the business success of a given airline is related to the efficiency of the proprietary airline system, known as AFDC (Airline Flight Dispatch), responsible for the control and the coordination between all airline functions such as flight operations, airport area operations, maintenance or system control. Notice also that some AOC messages might not be safety-related, and would more appropriately be called AAC (Aeronautical Administrative Communications) messages.³ As an example, the ENGINE (Engine Performance Reports) message is sent from the aircraft to the airline ground entity to provide a technical report about operational performances of engines and embedded avionic systems during the flight [12].

Aside operational services, airline companies are willing to enhance their offer to attract more passengers by providing new IFC (In Flight Connectivity) services (also known as APC—Airline Passenger Communications services). In fact, classic IFE (In Flight Entertainment) services, such as watching pre-recorded movies or listening to music stored on-board the aircraft are no longer sufficient for passengers. Modern customers ask for continuous broadband Internet access on-board the aircraft to share multimedia contents, send/receive e-mails, or connect to their favorite social network as they usually do at home. Also, additional airline applications may be supported in a long term perspective such as remote medical supervision or video surveillance systems to cope with a medical emergency or to monitor accidental/malicious events on-board the aircraft.

Alternatively to the terminology proposed by ICAO, a working group of ARINC (Aeronautical Radio, Incorporated) [14], a major provider of aviation communications and systems engineering solutions, a terminology for avionic network domains on the aircraft [15,16]. These four domains are:

- ACD (Aircraft Control Domain) regroups all systems dedicated to the control of the aircraft and the flight.
- AISD (Airline Information Service Domain) regroups non safety-related systems dedicated to the maintenance and the crew.
- PIESD (Passenger Information and Entertainment Service Domain) contains all systems allowing the passengers to access IFE/IFC services.

³ In some Refs. [13], AOC and AAC domains can be grouped under the AOC category since the same systems are usually used to send both AOC and AAC messages.

- PODD (Passenger Owned Devices Domain) is relevant to passenger owned systems which need be connected to the network (e.g. tablets, smart phones, laptops).

In order to successfully deploy digital data capabilities, the industry is likely taking some directions for the future deployment of these emerging data services.

2.5. Trends in aeronautical communications

This section looks at factors that could affect the future ATM infrastructure using data link systems.

2.5.1. Use of commercial off the shelf products

To take advantage of the fast pace of technological advances in IT industry, the aviation stakeholders are adopting COTS (Commercial-Off-The-Shelf) products and components for use in air-ground systems. There could be numerous potential benefits to use COTS technologies in aviation communications such as:

- COTS purchasing allows aviation acquisition to incorporate new technologies into data link systems more quickly than typical developmental programs.
- Reduction of research, development, implementation, and maintenance costs.
- Maturity of some existing COTS products.
- Seamless interoperability with existing terrestrial networks.

There is a new wave of COTS software and hardware vendors that understand the rigorous requirements of system certification and they are providing the avionics developers with true COTS solutions. Beeby [17] describes how the Seaweed systems [18] took the OpenGL standard graphics API (Application Programming Interface) and turns it into an avionics COTS product that can offer significant savings for systems developers.

When it comes to the ATN, introducing IP (Internet Protocol) to data link systems is often associated to the introduction of COTS. As it has been defined by ICAO, the deployment of an IP based ATN/IPS (Internet Protocol Suite) network will offer many advantages over the current ISO (International Organization for Standardization) OSI (Open Systems Interconnection) based implementation [19]. Besides, there is a myriad of IP-based COTS products that could be deployed at low cost as it has already been done for COTS software and hardware.

2.5.2. Use of satellite communications

With the paradigm moving progressively towards data communications, the satellite technology is already pointed as a suitable candidate for both operational and non-operational services. With the imminent use of data for safety-related aeronautical services, the introduction of new IFC/IFE passenger services, and the forecasted air traffic growth for future ATM environment, the aviation industry has so far shown a great interest in the use of satellite systems as supplementary means of air-ground communications. GEO (Geostationary) satellites are particularly an appealing solution due to their high performances and wide coverage. Currently L-band

SATCOM (Satellite Communication) technologies such as Inmarsat and Iridium are already used for safety related services.

Thanks to emerging Ku/Ka band satellite technologies, the use of satellite communications should be soon extended to passenger and airline services. Firstly, such high bandwidth satellite technologies allow airlines to improve their IFE solutions and provide new IFC services such as Internet on-board. Secondly, Ku/Ka satellite communications may play a great role for safety-related services as a complement to existing data link systems such as VHF, VDL (Very High Frequency DataLink) mode 2 [20], or L-band SATCOM. Indeed, communications between the pilot and the air traffic controllers must always be available upon demand and the air-ground channel availability must exceed 0.9995 in most cases, as mentioned by EUROCONTROL in [12]. Using satellite systems as a backup solution helps in reaching such a high availability demand. In past years, many projects tried to propose effective IFC solutions, using satellite technologies such as Boeing CCB (Connection By Boeing) [21], Panasonic eXConnect [22], or Airbus OnAir solution [23].

2.5.3. Integration of aircraft network domains

Current aeronautical communications are based on several segregated systems where each system is working independently over a dedicated link (HF, VDL mode 2, SATCOM etc.). Even though existing L-band satellite solutions are entirely dedicated to safety-related applications, it seems that using a single and seamless Ka/Ku band satellite solution which integrate all the aeronautical services is an attractive, cost-effective and long-term perspective solution: ATM RCPs (Required Communication Performance) such as link availability can be improved [24], and new passenger and AOC services can be afforded with decent performances. A plethora of R&D projects has shown the feasibility of an aeronautical satellite system able to integrate different communication technologies and domains into a global heterogeneous airborne network such as NEWSKY (Networking the Sky) [25], description = Seamless Aeronautical Networking through integration of Data links SANDRA (Seamless Aeronautical Networking through integration of Data links, Radios, and Antennas) [26], or ANASTASIA (Airborne New Advanced Satellite Techniques and Technologies in a System Integrated Approach) [27].

2.5.4. IP-based ATN networks

ICAO developed the ATN network as a strategy for integrating air/ground and ground/ground data communications into a global network serving ATC and AOC traffic. The ATN network is composed of IS (Intermediate Systems), ES (End Systems) and communication services which allow ground, air-to-ground and avionics data subnetworks to interoperate by adopting common interface services and protocols based on the ISO/OSI protocols [9]. The legacy ATN/OSI network offers a flexible, reliable, and robust communication system for future ATM communications thanks to two transport layer protocols specifically defined for the use in ATN end systems: the Connection Oriented Transport Protocol (COTP) and the Connectionless Transport Protocol (CLTP) [9]. These protocols

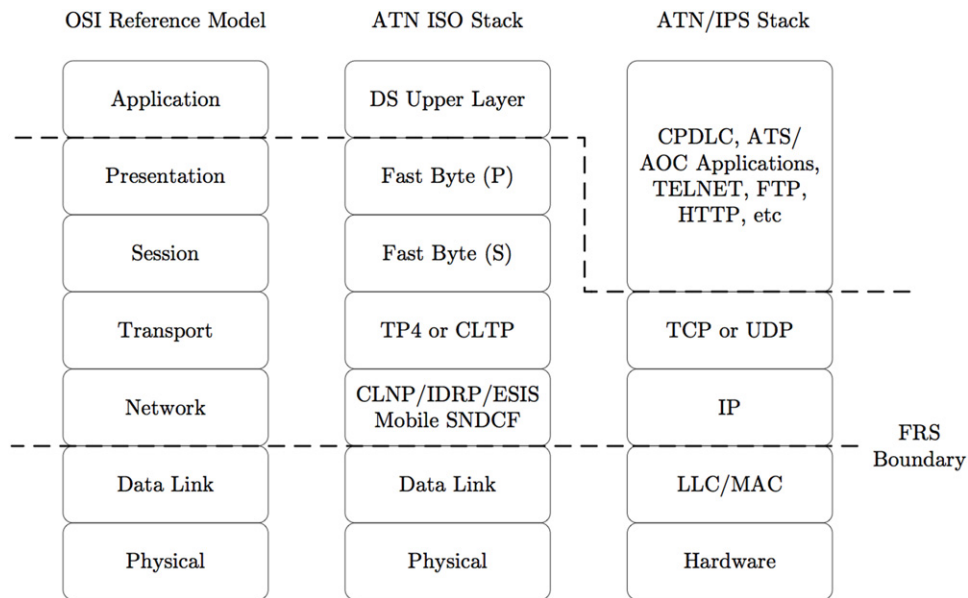


Fig. 2 – ATN/IPS protocol architecture.

are the equivalents of TCP (Transmission Communication Protocol) and UDP (User Datagram Protocol) in the IP-based networks.

Nevertheless, ICAO has also defined the ATN/IPS protocol stack as a potential next generation communication networks for future ATM in IP environment. Indeed, the ATN network will be likely used with existing IP networks having very heterogeneous characteristics (different delay, available bandwidth, packet error rates, etc.). Therefore, using an ATN/IPS protocol architecture allows reliable and seamless interoperability needed for such end-to-end IP communications. Fig. 2 shows the ATN/IPS protocol architecture as defined in the ICAO “Manual for the ATN using IPS Standards and Protocols” document [19].

The resulting aeronautical data link environment is shown in Fig. 3. Note that some technologies illustrated here (e.g. DVB, ADS-B) are discussed later in the paper.

While such a system solves many issues as discussed above and opens new perspectives for the future civil aviation, it has at the same time many drawbacks which must be considered, and network security is certainly on the top of the list. Benefits acquired from the use of data link technologies, COTS products, or IP connectivity, cannot be achieved without rigorous security considerations on the background.

3. Network security considerations

The impact of switching to data link systems is a substantial increase of the security risk in a critical context where human lives are involved and air-ground connectivity breakdowns cannot be tolerated. In aviation, security and safety are strongly tied. Both terms can be used differently depending on the considered context, therefore it is essential to primarily clarify their meanings in this paper:

- **Safety** is assimilated to the set of procedures, methodologies, and techniques deployed to mitigate risks related to the regulation and operation of the flight. For instance, one shall talk about safety-related or operational services for communications between the pilot and the air traffic controller;
- **Security** corresponds to the approaches and methods used to mitigate risks resulting from a malicious intent like an unauthorized intrusion on avionic systems. As a security attack may have some consequences on the regulation of the flight, security risks definitely imply safety risks in the aeronautical context. From a data link point of view, network security covers any attack or vulnerability in the air-ground communications.

This paper focuses only on network security issues related to aircraft communication technologies, referring to the “C” of the CNS paradigm concept. The strong relationship between safety and security in air transportation systems is further discussed in Section 7, however safety is not covered in this paper.

The civil aviation industry has shown in the past its interest in safety issues, but security should be considered as seriously as safety in the upcoming years. Indeed, the openness and heterogeneity of data link access technologies expose the ATN to malicious attacks, which emphasizes the security concern.

Scaling issues in ATM environments, mainly induced by growing air traffic load and demand for passenger flights, highlights the need for an optimized key management. The IATA (International Air Transport Association) reported in the 53rd edition of WATS (World Air Transport Statistics) a passenger growth of +22.1% between 1999 and 2008 [28]. As the number of aircraft/passengers/avionic systems using cryptography grows, it is clear that the amount of keys and digital certificates increases. Usually, PKIs (Public Key Infrastructure) [29] are used to cope with scaling issues in wired and

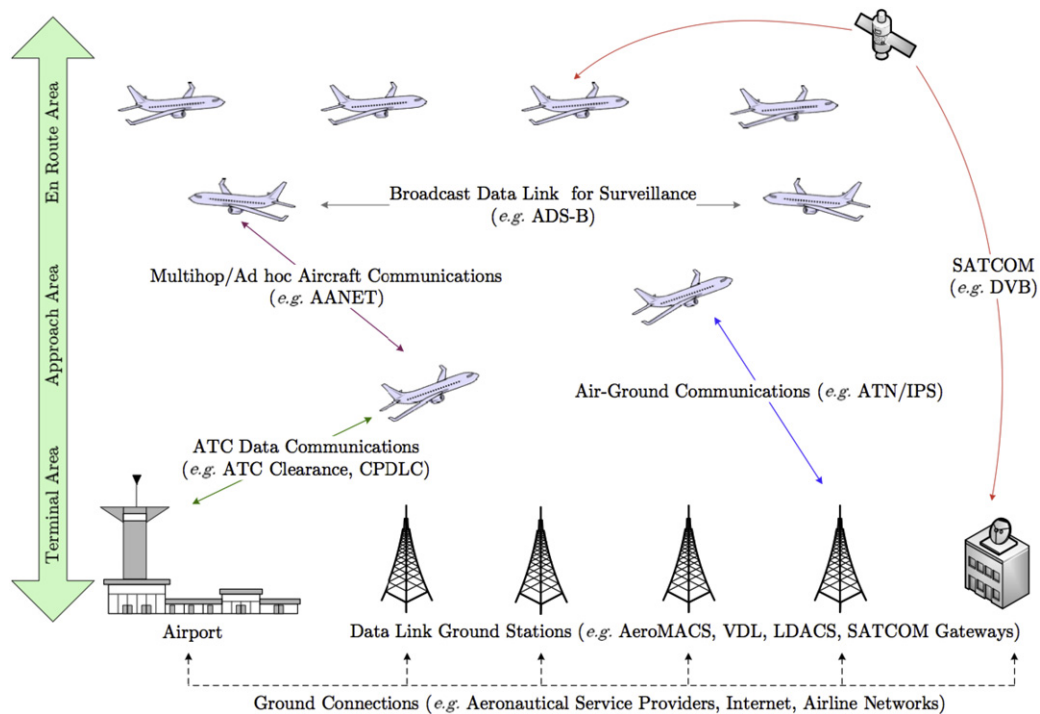


Fig. 3 – Data link communications in the future airspace system.

wireless networks and distribute the cryptographic credentials to end entities. However, such a solution should take into considerations the characteristics of the future ATN (large network, limited bandwidth, heterogeneity of services, etc.), and consequently should aim at inducing a minimal network overhead.

The opportunity to take advantage of safety-related service aggregation with airline and passenger services will require appropriate security countermeasures to protect the operational services from new threats. ATS traffic requires high priority and full availability, whereas AOC NG services may have restricting QoS (Quality of Service) requirements to be satisfied. While it is important for the aircraft to maintain continuous network connectivity with operational ground stations, availability becomes an important aspect of the overall network architecture, specially when considering the usage of a single satellite link for all the aeronautical services.

As usage and dependency on data link communications increase so do security risks. Security requirements for the communication system will be more complex to fulfill due to additional factors such as traffic heterogeneity, aircraft mobility or scaling issues. Providing IFC services for passengers may be an open invitation for hacking the air-ground connectivity as the Internet is an open network where anyone can connect anywhere no matter where they are. With broadband on-board connectivity, hackers may have much more opportunities to cause harm to data link communications.

Using COTS products heightens the potential security risks involved in air/ground communications. Alford [30] discusses the issues induced by the excessive use of COTS in aviation systems. Indeed, unlike proprietary security solutions, usually harder to break, COTS technologies are

widely used in terrestrial networks such as Internet, meaning that existing vulnerabilities are likely to be publicly known. The civil aviation should then use COTS carefully and a solid vulnerability survey shall be provided in order to track new threats and discover attacks before they get exploited by malicious users. For example, in order to fully benefit from IP technology capabilities for future ATN networks, it is important to understand existing security issues related to the protocol, to analyze how the research community managed to address them, and to consider the environment in which it is going to operate.

Fig. 4 shows a cause-effect diagram that illustrates conceptual relationships leading to network security concern in aeronautical data communications. Because of several factors such as economic conditions, bigger structures (e.g. airport, aircraft), and an increasing demand for passenger flights, the air traffic is continuously increasing. This air traffic growth, in addition to frequency spectrum limitations, results in a congestion of the frequency allocated to the civil aviation. For these reasons, the ATM and CNS paradigms need to be modernized and updated. Data link systems are the key to this renewal. Thanks to the maturity of COTS products, enhanced satellite technologies, and cost-effective network domain integration, data link communication is the future of the civil aviation. However, network security must be addressed in this case in order to avoid disclosure of important information and ensure a smooth and safe operation of the flight.

4. Basic security concepts

This section provides a brief introduction to generic security terms used in this paper: security requirements and basic cryptographic primitives are described.

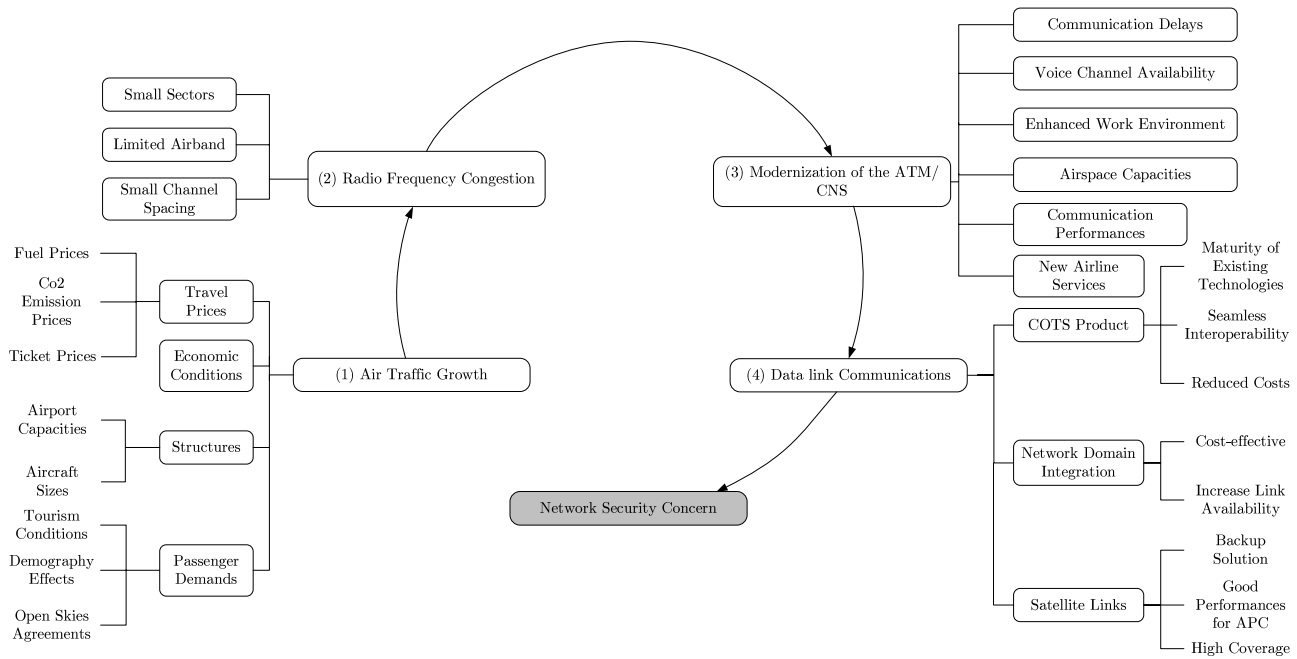


Fig. 4 – Cause-to-effect diagram for data link network security concern.

4.1. Security requirements

Usually, there are six main security services provided by secure communication systems. RFC 4949 [31] provides more details on security specifications for these services :

- **Confidentiality:** exchanged information are not disclosed to system entities unless they have been authorized to know the data. Confidentiality is achieved by encrypting the original message before sending, making it readable only by the intended receiver. Confidentiality is usually required when communications are classified top secret (e.g. military communications) or if data are considered sensitive (e.g. passwords, credit card identifiers, etc.). The privacy term could also refers to the confidentiality security service in the literature.
- **Authentication:** the sender or the receiver involved in the communication must be able to prove his identity when required by the other entity he is talking to. In this case, authentication is unidirectional (i.e. only one communicating part has to prove his identity). Otherwise, it is bidirectional (also referred by mutual authentication), meaning a strong authentication is required before initialising the transaction. When authentication is proven, malicious intruders cannot impersonate identities to mislead legitimate users.
- **Integrity:** data should always be amended by users who are authorized to do so. This security feature allows authorized users to verify that received information is identical to the one originally sent.
- **Non repudiation:** this security property speculates that communicating entities cannot deny the post or the reception of a message (i.e. providing a proof that an event actually occurred). Usually, non repudiation is supported by some cryptographic primitives such as digital certificates or signatures.

- **Availability:** network services and computer resources supplied by service providers must be always accessible and available to their customers when needed. In practice, this security feature is quite difficult to guarantee, specifically against DDoS (Distributed Denial of Service) attacks. DDoS attacks are intended to make these resources and services unavailable for a certain amount of time.
- **Authorization:** is the process of checking the privileges or permissions of an entity to access a system resource. Usually, authentication is required before authorization can be given to perform some actions on the system or the network.

Basically, these security features are necessary to ensure the security of the aircraft system. Such security requirements are essentially fulfilled using cryptography, which can be considered as the core component of any security protocol. As discussed in Section 7, security requirements may vary depending on the assets or the network domain to protect.

4.2. Cryptographic primitives

Fundamental cryptographic elements that can be used to provide security services are:

- **Symmetric keys:** in symmetric cryptography, end entities use the same key K to encrypt and decrypt a message M : $\{M\}_K K = M$. This symmetric key has to be shared and always kept secret between both entities. Symmetric keys are either exchanged across a secure channel or derived at both ends using specific methods such as the Diffie–Hellman algorithm [32]. Many algorithms have been proposed to provide symmetric cryptography such as DES (Data Encryption Standard) [33] and AES (Advanced Encryption Standard) [34].

- **Public and private keys:** in asymmetric cryptography, every end entity holds two keys; the public key is made publicly available to all other entities in the network while the private key is always kept secret. Asymmetric keys use one-way mathematical functions, which means it is considerably difficult to decrypt a message if it has been encrypted with one of the two keys. In other words, these keys are strongly related to each other. For example, if a message M is encrypted using the public key K_i^+ of an entity i , only the private key K_i^- allows to reveal the original message $\{M\}_{K_i^+}{}_{K_i^-} = M$. The reciprocal function remains true: if the message M is encrypted with the private key K_i^- , the public key K_i^+ is used to deduce the original message: $\{M\}_{K_i^-}{}_{K_i^+} = M$. RSA (Rivest, Shamir, Adleman) [35] is a widely used asymmetric algorithm in many security protocols dedicated to electronic transactions and even military applications. ARINC has specified the use of asymmetric cryptography in order to secure ACARS messages [36]. The document and ACARS security using asymmetric cryptography are discussed further in Section 6.2.1.
- **Digital certificates:** this is a central element in the use of asymmetric cryptography technique. A certificate is a data structure used to bind the public key K_i^+ to its owner i in an authentic way. The certificate has to be signed (see below) by a trusted third party called CA (Certificate Authority) and it ensures that the public key really belongs to the entity stated in the certificate: $\text{Certificate}(i) = \{i, K_i^+\}_{K_{CA}^+}$. A certificate aggregates many information such as a unique certificate number, the issuer identifier, the owner identifier, the public key, the cryptographic algorithm used to generate the signature or a validity period. Other fields can be included, depending on the certificate type or the purpose of use. The ITU-T (International Telecommunication Union-Telecommunication Standardization Sector) X.509 format is probably the most used certificate in Internet applications [37]. In civil aviation, the ATA (Air Transportation Association) Spec 42 document [38] defines the digital certificate profiles to be used in data link communications.
- **Digital hash functions:** this is a mathematical one-way function which takes a variable size data M and returns a fixed size value M' : $\text{Hash}(M) = M'$. A hash function output is called a hash value (also known as a checksum or digest). Hash functions have to be one way, meaning it is easy to compute M' if M is known, however if the digest M' is known, it is computationally hard to find M . Hash functions should be collision free, i.e. knowing M , it is computationally impossible to find $M' \neq M$ such as $\text{Hash}(M) = \text{Hash}(M')$. Finally the hash digest must always be fixed length, meaning if $M' \neq M$, $|\text{Hash}(M)| = |\text{Hash}(M')|$. SHA-1 (Secure Hash Algorithm) [39] is an example of a hash function which can be used to compute 160 bits length hashes.
- **Digital signatures:** a digital signature is the result of a cryptographic process used for user authentication and data integrity. A digital signature is produced as follow: a checksum is computed then encrypted using the private key K_i^- of the sender i : $\text{Signature}(i, M) = \{\text{Hash}(M)\}_{K_i^-}$. The resulting digital signature is then attached to the original data and sent. In order to verify the signature, the receiver decrypts the signature using the public key K_i^+ of the signer i , gets

the hash value, computes the hash of the original data and compares the two hashes: if they match then data integrity and sender authentication are proved.

The cryptographic primitives presented above allow to develop security countermeasures to the threats and attacks discussed in the following section.

5. Potential threats in data link networks

A threat is generally defined as any potential action that causes a damage to the network or the system assets. Usually the term encompasses both deliberate attacks (relevant to security) and accidental events (relevant to safety). In this paper, only threats resulting from intentional actions are considered. In order to provide a realistic list of potential threats in data link networks, assumptions have to be made for the intruder capabilities, this is usually called an intruder model.

5.1. An intruder model

The intruder model⁴ formally summarizes the attacker capabilities in the considered environment. Building a realistic and correct intruder model by taking into account the various aspects of the system and the environment is essential. In a computer network where messages are exchanged between several hosts, the Dolev & Yao intruder model [40] is usually used. An adversary is assumed to have the capabilities to capture, read, modify, create and replay messages exchanged between legitimate entities in the network. The Dolev & Yao intruder model assumes that the intruder does not have unlimited capabilities though, since it assumes that cryptanalysis is not possible to conduct (i.e. brute force or dictionary attacks).

In a data link network, the attacker has the same capabilities as he is able to intercept the wireless communications between the air and the ground entities. Formally, the intruder capabilities in a data link network can be summarized as the following:

- If the intruder knows messages M_1 and M_2 , he can constitute a new message $M_1|M_2$ where $|$ is the concatenation symbol;
- If the intruder knows M and the key K , he can forge and send $\{M\}_K$;
- The intruder is able to compute and use $\text{Hash}(M)$ if he knows Hash and M ;
- The intruder has the ability to generate Nonce numbers at will;
- The intruder is able to deduce M from $\{M\}_K$ if he has knowledge of K ;
- The intruder is able to deduce M from $\{M\}_{K_i^-}$ if he has knowledge of K_i^+ (and vice versa);
- The intruder is able to deduce M_1 from a message $M_1|M_2$ or $M_2|M_1$.

This intruder model will be helpful to establish a list of possible attacks on the data link network.

⁴ Also called adversary or attacker model in the literature.

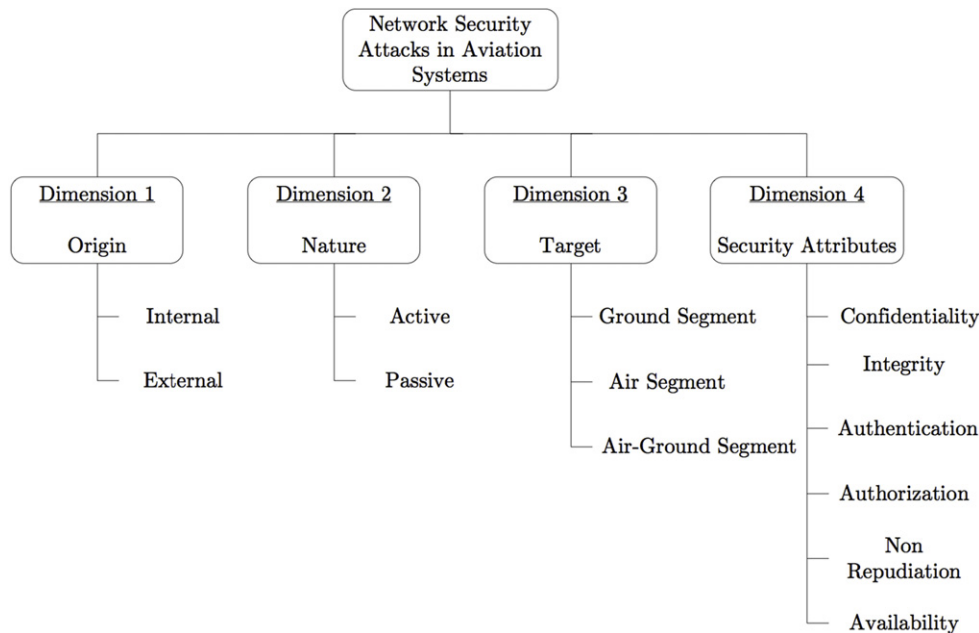


Fig. 5 – Network security attack taxonomy in data link systems.

5.2. A taxonomy of data link network potential attacks

An attack taxonomy is a classification means of existing or potential attacks. As attacks can behave differently, a data link network attack taxonomy is provided. The aim is to provide a useful classification scheme that can be easily shared, reused, and understood. The proposed taxonomy shown in Fig. 5 uses four different dimensions as explained below:

1. **The origin** is used to categorize attacks depending if they are internal or external:
 - *Internal attacks* are usually launched by a trusted entity (i.e. an ATC controller, or an airport technician). This trusted entity can conduct the attack voluntarily (e.g. a disgruntled employee who has been fired, vengeance, etc.) or it could have been compromised by a malicious attacker. However, internal attacks have a low probability of occurrence;
 - *External attacks* are launched by an entity that does not belong to the logical network or is not recognized as a legitimate actor of the system. External attacks are more likely to happen in data link networks.
2. **The nature** takes into account the purpose of the attack, it can be active or passive:
 - *Active attacks* directly interact with the target or the system (e.g. message injection, replay attacks);
 - *Passive attacks* are usually conducted by eavesdropping information that could be used later to launch an active attack. Due to the wireless nature of air-ground communication mediums, the ATM environment is prone to such attacks.
3. **The target** is one of the segments composing the FCI (Future Communication Infrastructure), namely:
 - *The air segment* which is the onboard system;

- *The air-ground segment* which is relevant to the communication medium (e.g. Satellite link);
- *The ground segment* which concerns ground networks.

4. **The Affected security attributes** correspond to the security services initially requested by the target but invalidated in the case of a successful attack. These security requirements are confidentiality, integrity, authentication, authorization, non repudiation, and availability as discussed in Section 4.1.

This attack taxonomy will be used in the Section 5.3 to categorize known and potential attacks in data link networks.

5.3. Known and potential attacks on data link networks

Generally, and by taking into account the intruder model capabilities described in Section 5.1, five potential attack categories on data link networks can be drawn:

1. **Eavesdropping attacks:** an attacker or an unauthorized user can listen without permission to the signaling/data traffic using specific decoders (e.g. ACARS) and network protocol analyzers (e.g. Wireshark);
2. **DoS attacks:** an attacker may flood the system with repeated messages (e.g. connection requests, TCP SYN flooding) in order to shut down the servers and make its services unavailable for a limited period of time. DoS attacks are also conducted using jamming technique by introducing a source of noise causing high interferences and reducing the channel capacity. For instance, the GBAS (Ground-Based Augmentation System) used to experience frequent shutdown in the Newark Liberty International Airport (New Jersey, U.S) [41]. As GBAS is used as a support for GPS (Global Positioning System) receivers of approaching aircraft, a federal investigation has been conducted to identify the root cause of the incident. It

has been shown that truckers, cruising along the turnpike next to the airport, have simply used GPS jammers to lure their bosses. While this act was clearly unintentional, it has emphasized the need for more secure communication infrastructure, at least in the airport perimeter;

3. **Masquerade Attacks:** an attacker can spoof/impersonate an authorized user identity to delude trusted communicating entities. For instance, pilots have a high level of trust in ATC clearances they receive via data link systems such as CPDLC. However, ATC clearances (as other CPDLC messages) are delivered using data link without authentication. This means that MITM (Man In The Middle) attacks are possible as soon as an intruder is able to impersonate the identity of an air traffic controller and send false ATC clearances. Getachew and Griner [42] provided an elliptic curve based authentication protocol for CPDLC, this security solution is discussed further in Section 6.2.2;
4. **Replay Attacks:** an attacker can intercept data link messages exchanged between two legitimate users and replay them when he needs to. For instance, in the original Needham-Schroeder public-key authentication protocol [43], an attacker is able to intercept a message exchanged between two entities involved in one protocol session, and replay it in a second protocol session in order to mislead both entities in a false authentication [44]. Such an attack can be imagined in unprotected data link protocols such as CPDLC or ACARS;
5. **Message Alteration Attacks** an attacker is able to delay, modify, or redirect messages exchanged between legitimate users. He can also inject messages he forged by respecting the message protocol format. For instance, an attacker could corrupt the contents of avionics software data and configuration files before its get loaded and installed on the aircraft.

Few attacks on data link communications have actually occurred in the past. In recent years, FAA networks and facilities were particularly vulnerable. In 1997, a hacker successfully gained unauthorized access to Bell Atlantic computer systems [45]. These systems provided critical safety services to the FAA control tower at the Worcester Airport (Massachusetts, U.S). The hacker had been able to shut down for six hours the tower main phone, some weather services, and a circuit which enables aircraft to send an electric signal to activate the runway lights on approach.

In 2006, a widely spread virus on the Internet affected ATC systems, forcing the FAA to partially shut down its traffic control facilities in Alaska. In 2008, nearly 800 incident alerts were reported to the ATO (Air Traffic Organization) responsible for ATC operations in the U.S. A report of the U.S DOT (Department of Transportation) said that 17% of those incidents have not been revised, while some of them could be critical as hackers may have taken control over ATO computers. In 2009, a group of Hackers compromised FAA public WEB application computers to gain unauthorized access to personal information on 48,000 FAA employees. Even if consequences were never really serious, such security incidents underlined the necessity to revise the network security policy of the FAA ground networks. Further details can be found in [46].

For the air-ground segment, attacks aiming to leak the ACARS messages exchanged between the pilot and the

ground entities do exist for over 10 years. Packet analyzers and software decoders are even freely available (e.g. <http://www.acarsd.org>) and allow the adversary to acquire a very large amount of information about the flight (e.g. flight level, destination, message content, ICAO aircraft code, etc.) as shown in Fig. 6. Referring to the taxonomy provided in Fig. 5, this attack has an external origin (i.e. unauthorized access to flight data), has a passive operation mode (the attacker simply eavesdrops the information without being detected, no harm is caused to the system directly), aims the air-ground segment, meaning the wireless data link medium, and finally threatens the confidentiality of the exchanged data. Table 1 classifies the attacks listed above according to the taxonomy provided in Section 5.2.

In the presence of such threats and advanced techniques, it is fundamental to provide security solutions which enhance the future data link networks.

6. Current state of data link network security

As described in [9], the data link network security requirements consist of access control, message integrity, and user authentication. Confidentiality has been most of the time considered optional [12]. The main identified threats are data alteration, message replay, and identity masquerading. In order to mitigate these threats and improve the robustness of the data link network, research projects and industry efforts are currently underway as shown below.

6.1. Overview of the industry initiatives

6.1.1. EUROCAE

The AISS (Aeronautical Information System Security) Working Group 72 of EUROCAE (European Organization for Civil Aviation Equipment) has been created to provide guidelines and methodologies to address security issues for aeronautical systems [47]. EUROCAE WG-72 is not intended to produce effective technical security solutions, but rather recommendations and guidance to support the growing use of highly integrated electronic systems and network technologies used on-board the aircraft. Within the scope of WG-72, EUROCAE is also studying the relationship between safety and security in civil aviation: existing safety assessment methodologies related to aircraft environment are constantly revisited in order to include security assessment (i.e. threats, vulnerabilities, attacks) in the global framework. Other EUROCAE working groups deal with security such as WG-82 where the AeroMACS (Aeronautical Mobile Airport Communication System) privacy sub-layer is being discussed [48].

6.1.2. RTCA

EUROCAE WG-72 is working jointly with RTCA SC-216 (Special Committee 216) to define MASPS (Minimum Aviation System Performance Standards) for aeronautical network security [49]. FIPS (Federal Information Processing Standards) standards provided by the NIST (National Institute of Standards and Technology) are included in MASPS for the American NAS infrastructure. RTCA SC-216 is also working on a

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acarsd 1.65 - (C).2003-2006 by KJM acarsd@acarsd.org 25 Jan 2011 - 17:18
Message label: H1 [Message to/from terminal] Block id: 1 Message number: F23A
Flight number: LA0704 Flight: Santiago de Chile, Chile-Frankfurt-Main, Germany via Madrid, Barajas,
Spain
Operator: Lan Chile
Message content: -
#M18POSN47101E008013,SUREP,161214,380,DITON,161415,KUDES,M54,32840,3
-----[ 25/01/2011 ]-----

1 contacts with this aircraft. First contact: 25/01/2011 17:09 as fl
ACARS mode: E Aircraft reg: CC-COF [A340-313 (X)]
Message label: H1 [Message to/from terminal] Block id: 4 Message nu
Flight number: LA0704 Flight: Santiago de Chile, Chile-Frankfurt-Ma
Spain
Operator: Lan Chile
Message content: -
#M18POSN47184E008206,DITON,161414,380,KUDES,161718,SNOM,M54,32334,3
-----[ 25/01/2011 ]-----

1 contacts with this aircraft. First contact: 25/01/2011 17:09 as fl

--=[ Good: 103, Upl: 0, Dupes: 21, Parity: 0, block: 0, crc: 0, Msgs: 124 ]--
--=[ CRC check: OFF ]--
Fetching informations for 'LA0704' from www.acarsd.org
Fetching flight informations for 'LA0704' from amadeus.net (In Queue 0)
Add Flight Routing LA0704 -> SCL-MAD-FRA to the OnlineServer
Found route 'SCL-MAD-FRA' for flightnumber 'LA0704' on amadeus.net
--=[ Press 'H' for help ]-- --=[ Flights: 36 ]-- --=[ Vol: 16 Extended: ON ]--

--=[ Daily Airlines statistics ]--
British Airways.....0007
General Aviation.....0007
Air France.....0006
Swiss International Air Lines/Crossair.....0004
Alitalia.....0003
Air Berlin.....0002
Condor.....0002
Royal Air Maroc.....0002
unknown.....0002
Cargolux.....0001
El Al.....0001
Emirates.....0001
Lan Chile.....0001

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Fig. 6 – Acars decoder user interface.

Table 1 – Data link network attack classification.

Attacks	Origin	Nature	Target	Security attributes
Eavesdropping	Both	Passive	3 Segments	Confidentiality
DoS	Both	Active	3 Segments	Availability Authorization
Masquerading	Both	Active	3 Segments	Authentication Non-Repudiation
Replay attacks	Both	Active	3 Segments	Integrity Availability
Message alteration	Both	Active	3 Segments	Integrity

series of documents to define security assurance and assessment processes for safety-related aircraft systems such as the DO-326/ED-202 “Airworthiness Security Process Specification” document [50]. Similarity to EUROCAE activities, security is addressed within other RTCA committees like SC-223 which develops standards for the airport wireless surface communications system [51]. EUROCAE WG-72 and RTCA SC-216 are currently defining two additional documents for airworthiness security: ED-203 [52] is about assurance through development practices and security countermeasures implementation on the aircrafts, ED-204 [53] provides worthiness guidance to assure that security countermeasures on the aircraft are working and maintained correctly.

6.1.3. ATA

The ATA (Air Transportation Association), also known as A4A (Airlines for America), created the DSWG (Digital Security Working Group) in order to provide recommendations and best practices regarding the use of PKI for data link security. DSWG provides industry specifications to facilitate the implementations of these information security practices and technologies. These recommendations are arranged in a document entitled “ATA Specification 42—Aviation Industry Standards for Digital Information Security” [38]. The ATA Spec 42 document deals with digital identity management and certificate profiles for the air transport industry.

6.1.4. JCG

The JCG (Joint Coordination Group) has been created in order to coordinate and harmonize all the computer and network security activities conducted by commercial air transport industrial actors. JCG is an organization of aviation professionals that attempt to motivate aviation standards development, primarily dealing with IT and network security. The claimed goal of the JCG group is not to produce any kind of security standards, but rather give a global overview and centralizes the ideas and point of interests related to the civil air transport security community.

6.1.5. AEEC and ARINC

The AEEC (Airlines Electronic Engineering Committee) organization produces engineering standards and technical solutions for onboard systems (e.g. embedded networks, avionics, cabin systems, etc.). AEEC develops and adopts the so called ARINC standards, such as ACARS for instance. AEEC created several subcommittees dealing with information security interests, namely NIS (Network Infrastructure and Security) and DSEC (Data link Security).

The NIS subcommittee aim is to coordinate the standardization of IP connectivity and security to the aircraft. NIS is some kind of a discussion forum about network and security issues, providing a better coordination between subcommittees working on other aspects of aircraft and non-aircraft systems. The claimed goal is to avoid redundant activities,

support interoperable solutions, and provide operational assessments of networking and security issues. Currently, the NIS subcommittee is working on the digital certificate usage in the aircraft environment, specially to provide guidance to aircraft manufacturers and operators about the use of PKI standards for example. NIS subcommittee coordinates permanently with the ATA DSWG to avoid duplicate provisions as both are updating security provisions in the ATA Spec 42 document.

The DSEC subcommittee primary goal is to develop a standard called ARINC 823 (also known as AMS—ACARS Message Security) to be used as a basis for design and implementation of security mechanisms for ACARS communications. The standard has been defined in a two-part document: part 1 [54] contains a specification of the AMS security protocol, and part 2 [36] specifies life-cycle management of the cryptographic keys necessary to secure AMS operations. Another AEEC contribution to network security in civil aviation is the “Commercial Aircraft Information Security Concepts of Operation and Process Framework” document [55]. Also known as ARINC Report 811, the document provides additional guidances to deal with both physical and operational security considerations in commercial aircraft. The risk-based framework is assumed to facilitate the development of cost-effective aircraft information security and provide a common language for understanding security needs.

6.1.6. ICAO

ICAO ACPs (Aeronautical Communication Panels) were created in 2003 in order to develop standards and recommended practices for data link aeronautical communications. Among the four working groups created by ICAO, WG-I is probably the most relevant to security concerns. ICAO WG-I primary goal is to develop guidances to use IPS in future ATN applications. IPS security has been largely addressed for air-ground and ground-ground communications. Several guidance to use and implement IPsec (IP Security) [56] and IKE (Internet Key Exchange) [57] protocols have been provided in the scope of WG-I meetings as stated in [58]. IP mobile security is also addressed [59]. All these guidelines are meant to update the security requirements listed in the ICAO “Manual for the ATN using IPS Standards and Protocols” document [19] as indicated in [60]. Also, chapter 13 of the Annex 17 document [61] has been recently amended to include information security. A risk assessment methodology has been developed within the ICAO threat and risk working group to embrace network security threats.

6.1.7. EUROCONTROL and FAA

The European collaborative project entitled SESAR (Single European Sky ATM Research) aims to modernize the future European ATM by offering technical and operational solutions to meet future air traffic capacity and air safety needs [62]. A similar project called NEXTGEN (Next Generation Air Transportation System) is undertaken by the FAA (Federal Aviation Administration) to improve the American NAS (National Airspace System) and deal with the same air traffic congestion issues [63].

Both EUROCONTROL and FAA focus on security aspects in SESAR and NextGen. The 16th SESAR WP is a transversal

area working package meant to cover common requirements for the rest of WPs. Among other common tasks, three security areas are covered, namely airspace security, ATM security standards, and CNS security (e.g. ADS-B security, ATN security, etc.). Koelle and Hawley [64] provides a general overview of security management in SESAR. The FAA NEXTGEN project considers also security as an element inherent in all aspects of NAS operations. Several security aspects are currently covered by NEXTGEN including security threat detection, tracking, and secure integration of heterogeneous information systems.

6.1.8. European commission

Toward the modernization of the ATM and CNS paradigms, the EC (European Commission) funded three major research projects, namely STAR (Secure ATM CDMA Software Defined Radio) [65], SANDRA [26], and NEWSKY [25]. These projects deals with security issues such as service integration through a single data link, IP security (e.g. routing security, IP mobility), threat analysis, or key management.

6.1.9. CEN TC 377

The CEN (European Committee for Standardization) is collaborating with the EUROCAE in the field of ATM. The CEN TC 377 working group has produced an extension of the ISO 27001 standard and adapted to the civil aviation [66]. The resulting document entitled “Air Traffic Management—Information Security for Organizations Supporting Civil Aviation Operations” specifies the security requirements and measures of an information security management system in the scope of civil aviation [67].

6.1.10. ECAC

The ECAC (European Civil Aviation Conference) [68] is an organization of 44 European member states and an integral part of the ICAO global air transport family. Among others, ECAC deals with network and IT security in civil aviation throughout a study group which writes guidance on security control measures.

Table 2 summarizes the industry and research initiatives towards IT and network security issues.

6.2. State of the art of data link network security proposals

The industry and research initiatives depicted in previous sections provided several security mechanisms and protocols to secure data link communications. In this section, a critical review of the relevant literature is provided.

6.2.1. ACARS message security

ARINC provided the AMS framework for ACARS which defines a set of functions shared among an aircraft, a data link communications provider, and an airline to execute processes and protocols that result in protecting ACARS messages [54,36]. AMS uses the existing ACARS infrastructure. For interoperability matters, the ACARS security solution has been implemented at the application layer: a SAM (Secure ACARS Manager) application, located on the CMU

Table 2 – Summary of industry initiatives as regards to IT and network security.

WGs & Projects	Security concerns	References
EUROCAE WG-72	Development of security guidelines	[47,50,52,53]
EUROCAE WG-78	AeroMACS security	[48]
RTCA SC-216	Security airworthiness	[49,50,52,53]
RTCA SC-223	AeroMACS and networking security	[51]
ATA JCG	Coordination of security activities	[69]
ATA DSWG	PKI interoperability, best practices	[38]
AEEC NIS SC	IP connectivity standardization	[70]
AEEC DSEC SC	ACARS security, ATN security	[36,54]
ICAO ACP WG-I	IPS security	[71]
ICAO ACP WG-M	Maintenance security	[72]
ICAO ACP WG-S	AeroMACS security	[73]
SESAR WP 14	SWIM security	[74]
SESAR WP 15	CNS Security	[74]
SESAR WP 16	Security risk management	[64]
NEXTGEN	ATN and SWIM security	[63]
EC STAR WP 1	Risk analysis, security requirements	[65]
EC SANDRA	Integration of IP data link network	[26]
EC NEWSKY	Threat analysis, IP security	[25]
CEN TC 377	Security requirements	[66,67]
ECAC	Security control measures	[68]

(Communication Management Unit) avionics, provides all the required security features to establish and maintain a secure connection between the peers. The same SAM application is implemented on the ground airline operation centers for an end-to-end security solution. AMS uses a key establishment algorithm to create a session key for each secure connection between the peers. This key establishment scheme uses elliptic curve signature for authentication and integrity, and certificate repository for identity management.

In order to preserve bandwidth (e.g. cryptographic operation overhead), AMS uses the Deflate compression algorithm specified by the ATN SARPs, which reduces the volume of airline and air traffic data by a factor of three to five. The aircraft identity, bound to the AMS header, is not considered as a sensitive information in civil aviation. However, in military and private aircraft, it should be protected. For this purpose, it is encrypted in order to provide aircraft identity confidentiality.

6.2.2. Secure CPDLC

Getachew and Griner [42] investigated an elliptic curve-based authentication protocol for CPDLC communication systems. Mutual authentication between the pilot and ATC ground systems have been provided in order to avoid identity masquerading and spoofing attacks. In order to establish the CPDLC data communications between a controller and a pilot, a CM (Connection Management) step is required. It consists of a CM-logon services which allow a CM air user (e.g. pilot) to initiate a data link service and provides information on each data link application. If the CM-logon is successful, the CM ground user (e.g. a controller) responds to the air user by indicating which data link application it can support. This step can be seen as a pre-establishment of CPDLC connection. Once the connection established, both air and ground users are allowed to exchange CPDLC text-based messages.

The solution provided in [42] is divided in two parts: for each CM-logon, the communicating entities have to establish a unique application session key. To generate the session key,

both GCM (Ground Connection Management) and ACM (Air Connection Management) entities must calculate a shared public value $X_{GCM,ACM}$ during the CM-logon. An elliptic secret value derivation primitive SV and a KDF (Key Derivation Function) are used to compute $X_{GCM,ACM}$. $X_{GCM,ACM}$ is then used to calculate a session key $K_{ACP,GCP}$ between the ACP (Aircraft CPDLC) and the GCP (Ground CPDLC) entities. For authentication purposes, a HMAC (Hash-based Message Authentication Code) [75] $MAC_{GCP,ACP}$ is computed and sent with a CPDLC data message M_{GCP} . At reception, the aircraft CPDLC entity calculates $MAC'_{GCP,ACP}$ using M_{GCP} and compares it to the original $MAC_{GCP,ACP}$. The proposed protocol uses also a PKI to exchanges certificates between involved entities.

Getachew and Griner [42] provided a promising approach to secure the CPDLC application, however there are some issues that should be addressed. First, the public/private key pairs generation and exchange provided by the authors is vulnerable to MITM attacks. The original procedure works as the following:

1. $(K_{CA}^-, K_{CA}^+) \leftarrow KGF_{CA}(T_{CA})$ // KGF is a Key Generation Function
2. $(K_i^-, K_i^+) \leftarrow KGF_i(T_i)$ // T_i is an Elliptic curve domain parameter
3. $i \rightarrow CA : [K_i^+, ID_i]$ // ID_i is a unique identifier of entity i
4. $CA \rightarrow i : [K_{CA}^+, ID_{CA}]$ // CA sends its public Key to i
5. $CA \leftrightarrow Certificate(i)$ // CA generates the certificate for i

An attacker is able to generate its own public/private keys (at message 2) and make the CA generates a valid certificate for later use (message 5). Also, the attacker is able to impersonate the CA and generate a false certificate for the entity i . An identity-based key management which removes the need for public key queries or certificates could be helpful to mitigate such weaknesses. Besides, it was found that the shared secret generation process of this proposal lacks clarity. Indeed, the shared secret is derived as $Z_{ACM,GCM} \leftarrow$

$SV(K_{GCM}^+, K_{ACM}^-)$ from the aircraft side, and $Z_{GCM,ACM} \leftarrow SV(K_{ACM}^+, K_{GCM}^-)$ from the ground side. This means that implicitly, $Z_{ACM,GCM}$ is equal to $Z_{GCM,ACM}$ whereas the SV function uses different inputs. This part of the protocol seems unclear and misleading.

Another paper focused on the security of CPDLC over ATN: McParland et al. [76] presented a set of cryptographic techniques in order to improve the overall security of pilot-controller communications. For the specific key management and agreement schemes, they suggested the use of a dedicated ATN PKI. This is slightly the same as the one presented by Getachew and Griner [42]. The authors underlined the fact that the ATN schemes described in their paper have been placed in the ATN upper layer services and accordingly are available to any of the ATSC applications (e.g. FIS—Flight Information Services).

Olive [77] presented an overview of the ATN security concept. Rather than presenting a technical solution to mitigate security risks related to ATN applications, the authors discussed the cryptographic techniques chosen in the ATN. In order to minimize security data overhead in the bandwidth-limited aeronautical communications environment, the authors presented several optimization techniques such as compressed X.509 certificates, elliptic curve cryptography, and hybrid key agreement (i.e. both asymmetric and symmetric schemes are combined).

6.2.3. AADS and EDS security

Robinson et al. [78] presented a security framework for a specific aeronautical network application, namely the EDS (Electronic Distribution of Software). The EDS application aims to distribute information assets such as software and data when the aircraft is in maintenance, in production, or on ground at the terminal. As these software are to be used when the aircraft takes-off, it is essential to ensure the integrity and authenticity of information loaded into the aircraft. Thus, Robinson et al. [78] tried to identify main security threats targeting the EDS application, then they proposed a secure EDS system called AADS (Airplane Assets Distribution System), which addressed these threats and served as a guideline for later EDS design and implementation. The AADS system used digital signatures and key management and distribution schemes. As an extension of this work, Robinson et al. [79] presented two security approaches for their AADS system: an ad-hoc technique without trust chains between certificates, and a structured approach employing a third party PKI for EDS on commercial airplanes and based on the CC (Common Criteria) standard [80]. The provided protocol is a hop-by-hop signature mechanism between a software supplier, a software owner, an aircraft, and a certificate authority. The authors suggested that a part of the software is signed, sent hop by hop then verified until the aircraft constitutes the complete piece of the software. In our opinion, this approach affords a higher level of security but it is lowly scalable in the case there are many hops and many part of softwares.

Oheimb et al. [81] provided a security assessment of the AADS by combining the CC methodology and the AVISPA (Automated Validation of Internet Security Protocols and Applications) formal model checking tool. An AADS generic

component called ASV (Asset Signer Verifier) has been introduced. Basically, the AADS is a chain of ASVs instances running on each node of the architecture. ASV protection profiles have been established using CC then the AADS high level protocol has been formally specified and tested to check if the security requirements have been respected or not.

These proposals can be effective in order to improve the security of each application. However, data link security at the application layer may suffer from several weaknesses. First, these solutions are relevant to cockpit communications and specific applications, which usually requires enhancements and modification on an application basis. Besides, critical issues such as services priorities and non-operational communication security remain unsolved. Furthermore, neither interoperability nor scalability issues can be addressed. PKI systems have been mentioned as potential security solutions without a real assessment or adaptation to the specific aeronautical context. In Section 8, a PKI to deal with such issues is provided.

6.2.4. Transport layer security

Providing security at the transport layer has one advantage over the security at the application layer, as it does not mandate modifications to each application and provides a transparent security for users. SSL (Secure Socket Layer), also known as TLS (Transport Layer Security) [82] has been discussed in several studies as an alternative to application-based security [83,84]. However, transport layer security has several drawbacks that makes it inefficient in some circumstances. For example, TLS is only able to secure TCP flows and does not provide any security mechanisms for UDP flows [85]. Indeed, TLS needs to maintain context for a connection and is not implemented over UDP. This could be inconvenient if real-time applications, such as telemedicine or video surveillance systems that likely use UDP instead of TCP, are considered. In this case, network layer security could be an alternative to secure upper packet layers without any considerations to the nature of transport layer flows (i.e. TCP or UDP).

6.2.5. IP security

Another aspect that has been covered by the aviation community is the IP network connectivity between the aircraft and the ground stations. Ehammer et al. [86] provided an overview of IP-based threats against aeronautical networks. They focused mainly on network logical separation using network tunnels. In [87], Ali et al. [87] depicted the scalability issues related to the use of IPsec in the scope of ATN networks. They revealed the computational high consumption related to various encryption and decryption processes within IPsec. Then they proposed a backup solution, namely an anomaly detection engine within an NIDS (Network Intrusion Detection System) in order to monitor malicious acts within the operational and non-operational network domains. The anomaly-based IDS uses statistical method to detect unusual behaviors in the avionic network.

IP multicast can be used to provide IP services to passengers via a satellite link. In such a case, the ANSP (Aeronautical Network Service Provider) may want to control the access to

multicast sessions and account the usage of those multicast services for charging the passengers. For this purpose, Pillai and Hu [88] presented a AAA (Authentication Authorization and Accounting) based security framework which supports both post-paid/pre-paid passenger accounting and multicast session monitoring. The MSEC-AAA framework combines the GDOI (Group Domain OF Interpretation) IETF multicast security architecture with classical AAA features such as authentication and billing. The GDOI protocol is set between a group member and a group controller for SA (Security Association) establishment. The system architecture is composed of an aircraft network and a ground network. A Diameter [89] AAA server has been deployed on each of them: on the aircraft network, the AAA server provides proxy AAA functionalities for passengers whereas on the ground network, the AAA server provides authentication and authorization for access to multicast services and also accounting and billing functionalities. The presented AAA framework is composed of three phases: passenger authentication by the aircraft network, passenger authentication by the content provider network which sets up the security keys needed to protect the multicast services. Finally, a rekeying phase has been defined in the case security keys are changed. [90,91] extended their work on MSEC-AAA framework by using GSAKMP and EAP as a means to protect GDOI messages. In [90], several EAP methods have been tested (e.g. EAP GPSK, EAP SAKE, EAP PSK, and EP SKL) with a focus on time delays induced by EAP message exchanges.

6.2.6. Satellite communication security

IPSec should play a major role in data link network security as it has been quoted in the ICAO ACPs. However, the end-to-end protocol is unadapted to satellite communications if PEPs (Performance Enhancing Proxies) [92] are used. Indeed, PEPs can be deployed in order to improve the performances of satellite links suffering from high latency issues. As PEP agents need to access the TCP header, IPSec fails in delivering such information because the TCP flow is encapsulated then encrypted at the network layer. This issue has been identified in [93] by Thanthy et al. [93] whom suggested the use of SSL/TLS-based security mechanism for aircraft data networks. ML-IPSec (Multi-Layer IPSec) resolves the conflict between end-to-end security in standard IPSec and working of PEPs. Cruickshank et al. [94] presented the concept and detailed design of ML-IPSec by breaking the IP datagram into three zones while enabling the intermediate nodes to access the TCP header and HTTP header information. Luglio and Roseti et al. [95] addressed ML-IPSec and PEP compatibility issues then assessed the performances of the network through simulation under different network security schemes.

Currently, there is no satellite security framework that has been developed specifically for aeronautical communications. Nevertheless, a lot of work has been done in the scope of IP-based satellite communications. In [96], an overview of existing threats on IP-based satellite communications has been presented. The authors made a security analysis of these threats and provided a set of security requirements for IP over satellite DVB (Digital Video Broadcasting) networks. SatIPSec (Satellite IPSec) [97], which is a derived version of the IPSec protocol for satellite communications, has been proposed in the scope of the SATSIX (Satellite-based Communications

Systems within IPv6 Networks) project. SatIPSec is similar to IPSec while it addresses the PEP compatibility issue of the original IPSec protocol.

The DVB-RCS (Digital Video Broadcasting-Return Channel via Satellite) [98], which is currently a standard proposed by the ETSI (European Telecommunications Standards Institute), can be used for providing IP based services along with audio and video broadcasting. For this purpose, a new protocol, called ULE (Unidirectional Lightweight Encapsulation) has been defined by the IETF (Internet Engineering Task Force) for the transport of IP packets over DVB. ULE provides several benefits over the existing mechanisms used in DVB for transporting IP data. Several solutions have been provided in order to secure IP over DVB and ULE protocols. Cruickshank et al. [99] presented an ULE security solution using ULE extension headers. Noisternig et al. [100] extended their work by presenting the ULE security header format together with processing sequence in both transmitter and receiver sides. Pillai and Hu [101] also specified an extension to the ULE protocol in order to provide security for IP services over DVB. The different security requirements for IP over DVB services and the security extensions for ULE have been discussed. The impact of these security mechanisms on the IP traffic through has been also addressed. Similarly, Salam et al. [102] proposed an enhanced security framework for DVB-RCS security specifications adopting ULE encapsulation to provide secure lightweight encapsulation. Cruickshank et al. [103] provided a solution to secure multicast services over DVB-RCS.

6.2.7. AANET security

Currently, several work are being conducted in the field of AANETs (Aeronautical Ad hoc Networks). Their feasibility on both continental and transatlantic aeronautical areas has already been demonstrated [104–106]. AANETs are specific mobile ad hoc networks where an e-enabled aircraft acts as a self-aware node and communicates with other aircraft and ground entities. These networks can be used to provide a myriad of aeronautical or passenger applications such as Internet connection, operational services (e.g. black box data downloads) or more specific AOC services. As for mobile and vehicular ad hoc networks (MANETs, VANETs), these emerging network systems require specific routing protocols to cope with the aeronautical environment constraints.

In this line, several AANET routing protocols have been provided recently such as GRAA (Geographic Routing Protocol for Aircraft Ad Hoc Network) [107], AeroRP (Geographical Routing Protocol for Highly Dynamic Aeronautical Networks) [108], or ARPAM (Ad-hoc Routing Protocol for Aeronautical Mobile Ad-Hoc Networks) [109]. These protocols suffer from a blatant lack of security making the AANETs defenseless against network attacks such as select forwarding, Sybil attacks, sinkhole or spoofing attacks. On the other hand, there is a strong need for confidentiality in inter-airline communications (e.g. Kerosene consumption, passenger lists, intellectual property, etc.) when packets are routed from the sender to the receiver. If intermediate nodes (i.e. aircraft) do not belong to the same airline as the destination node, they will be able to decrypt the packets while they have not the

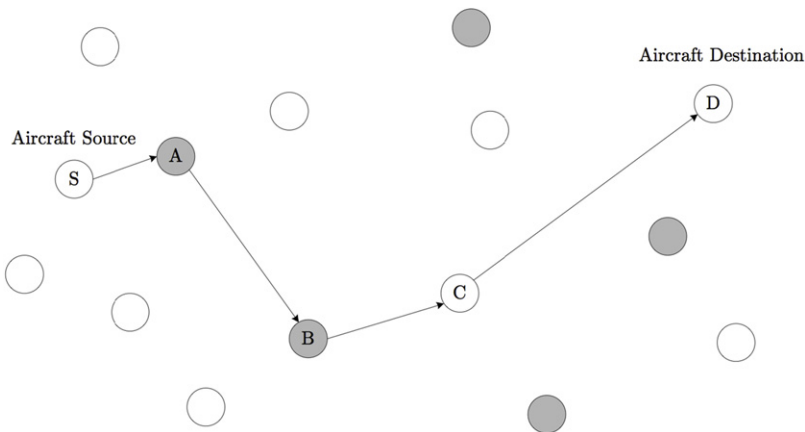


Fig. 7 – Illustrative case of need for confidentiality in AANETs.

authorization to do so. Fig. 7 shows a scenario where the aircraft source S sends an AOC packet to the aircraft destination D. If the routing protocol picks a path such as the packet goes through aircraft belonging to other airlines (namely nodes A and B), the latter will be able to read the content of the packet if it was not encrypted.

Sampigethaya et al. [110] provided a security analysis of AANETs. The authors proposed digital signature as a solution building block for AANET security providing data integrity. Nevertheless, digital signature is not sufficient to cover the inter-airline confidentiality issues mentioned above. Iordanakis and Dilintas [111] presented a more in-depth security analysis of AANET routing issues. Iordanakis and Dilintas [111] focused on the ARPAM routing protocol. They made a comprehensive classification of potential threats on ARPAM (e.g. message tampering, wormhole attacks, Byzantine attacks, selective forwarding). Despite a technical review of potential vulnerabilities in the ARPAM routing protocol, the authors did not mention any solution to address such security issues.

6.2.8. Ground interconnection security

Karmarkar and Martin [112] provided several guidance to mitigate potential distributed network attacks on ground networks outside the ANSP (Air Navigation Service Provider) domain. For instance, in order to cope with network attacks via CSP (Communication Service Provider) and non-ASNP interfaces (e.g. DoS attacks), the authors suggest the use of rate limiting through source-based or destination-based metering⁵ techniques. However, the false positive ratio can be very high when traffic metering is used, which prevent legitimate data transfers between the CSP and non-ASNP networks. Access lists are also suggested as DoS countermeasures to ensure that pair-wise communications can only happen between authorized entities. However, access lists can be very heavy and need to be retrieved from a access list repository. Their transfers should also be protected against alteration and MITM attacks.

⁵ Metering consists of limiting the number of exchanged messages based on pre-established normal traffic profiles for a given application.

6.2.9. AeroMACS security

AeroMACS has been identified as C-band technology candidate that suits the best for the provision of dedicated aeronautical communication services on the airport surface. As the AeroMACS is based on the IEEE 802.16e/802.16-2009 standard, it logically uses the same link layer security mechanisms of the WiMAX (Worldwide Interoperability for Microwave Access) IEEE 802.16 protocol [113] technology. A privacy sub-layer has been defined to deal with several attacks targeting the integrity, authorization, and confidentiality of exchanged data. The central component of the AeroMACS security is the PKM (Privacy Key Management) framework which is responsible for nearly all the security operations in the AeroMACS network (e.g. key generation, base stations and mobile stations authorization and authentication, etc.) [114]. Despite these security mechanisms, several security weaknesses still exist. The MOPS (Minimum Operational Performance Specifications) of the AeroMACS is currently being written, then such vulnerabilities should be corrected before the final release of the standard. These weaknesses (mainly derived from those already known on WiMAX) could be categorized as the following:

- *Unauthenticated management messages*: most of the management messages defined in AeroMACS are integrity protected using a HMAC or alternatively a CMAC (Cipher-based Message Authentication Code) [115]. However, some messages (e.g. MOB_TRF-IND, MOB_NBR-ADV, RNG-REQ) are not authenticated which leads to some vulnerabilities. Also, some management messages are sent over the broadcast management channel: authenticating this type of messages becomes difficult since there is no common key to generate the message digest. Furthermore, a common key would not completely provide message integrity as mobile stations sharing the same key can arbitrary forge these management messages and generate false authentication digests;
- *Unencrypted management messages*: when initial network entry procedure begins between a MS and a BS, many control information (e.g. mobility parameters, power settings, security capabilities) are sent in clear. An adversary may eavesdrop these messages just by listening to the channel in order to establish detailed profiles of MS or BS. More

Table 3 – Data link security proposals.

Security mechanism	Protocol layer	Comments
AMS [36,54]	Application	– For ACARS only – Already obsolete
Authentication Protocol for CPDLC [42,76]	Application	– MITM attacks – Misleading key management
ADS Signature [78,79,81]	Application	– Low scalability
SSL/TLS [83,84]	Transport	– Only for TCP Flows
IPSec [87]	Network	★ PEP incompatibility [94,95] ★ High data overhead [116,117]
Anomaly-based NIDS [87]	Network	– Based on statistical methods
MSEC-AAA [88]	Network	★ Unsigned GDOI messages [90,91]
ARPAM [111]	Network	– Insecure routing scheme
ANSP traffic metering	Network	– False positive ratio
Access lists [112]		– ACL repository needed – Unprotected ACL transfers
AeroMACS [114]	Data link ^a	★ Unsigned control messages [118] ★ Unencrypted control messages [118] ★ BS rogue attacks in multicast [118]
DVB security [98]	Data link	★ Unsecure multicast services over DVB-RCS [96,103]

(★) Symbol means the security issue has been addressed in the reference afterward.
^aData link in this table refers to layer 2 of the protocol stack.

specifically, when MBRA (Multicast and Broadcast Rekeying Algorithm) is used, group encryption keys called GTEKs (Group Traffic Encryption Keys) are distributed to all group members and encrypted using another key, namely GKEK (Group Key Encryption Key). This GKEK symmetric key is shared and known by all groups which means that a malicious group member may use a new GKEK key to update request for the GTEK key and generate its own GTEK;

- *Weak key management in the multicast and broadcast service mode:* multicast and broadcast messages are encrypted and authenticated using a symmetric shared key between a BS and all MSs belonging to the same group: this is an issue in the sense that any MS may impersonate the original BS by forging false multicast or broadcast messages and sending them to other MSs in the same group.

The multicast and broadcast group services of AeroMACS are currently being discussed in order to define if they should be included in the MOPS. Still, other security breaches need to be avoided: the security of AeroMACS extends beyond the privacy sublayer. AeroMACS inter-networking security issues should also be discussed (e.g. handover, DHCP, mobile IP, AAA security).

Table 3 summarizes the data link security proposals and the associated security weaknesses or functional issues discussed in this section. In some cases, there have been solutions to the security issues mentioned in the table. References to these works are also provided.

7. Security challenges in data link networks

This section summarizes what should be the main points of interest regarding cyber security in next generation aeronautical communications.

7.1. Data flow logical separation

Currently, ATS communications have to be strictly separated from other types of communications, because of safety and regulation policies, as required by ICAO SARPs (Standards and Recommended Practices) [24]. ICAO SARPs define a set of end-to-end protocols and operational access procedures that allow both safety and non-safety aeronautical applications to use data link technologies independently of air-ground and ground-ground sub-networks.

However, as explained in Section 2.5.3, increasing network integration is encouraged to improve the network efficiency and provide a better use of the available resources and lower the costs. Several possibilities to integrate operational with non-operational services on different protocol layers have already been studied in the scope of the NEWSKY [119] and ANASTASIA projects [120].

In the perspective of a network architecture which allows the coexistence of all aeronautical services in the same infrastructure, an exciting challenge from a security point of view emerges. While the separation on RF frequencies, as performed nowadays, offers the highest level of separation and safety, other options for separation exist within other layers such as link (e.g. using PDUs) or network layers (e.g. IPSec).

Indeed, the security architecture should, not only satisfy different security requirements for each traffic class (i.e. data flows belonging to a network domain), but also provide a logical and efficient segregation of ATS services in order to be fully compliant with ICAO SARPs. Besides, efficient QoS policies (e.g. traffic shaping, priority queuing) have to be designed for a preferred treatment of operational data as compared with non-operational data. This is a key point to prevent an excessive bandwidth consumption by non-operational applications when the network or system link capacity does no more provide enough resources for all the

applications, which can be considered as a critical security issue. Secure tunnels can be used, not only to ensure data protection and authenticity, but also guarantee the separation between operational and non-operational data.

7.2. Heterogeneity of security requirements

In operational aeronautical communications, some security properties are more or less required than others. For instance, integrity and availability need to be highly provided for ATS services as these are vital for the safety of the flight. In this case, confidentiality is not mandatory as carried ATS data are not really sensitive. However, it could be important for airline companies to protect some of their AOC data from unauthorized access for competitive matters and prevent themselves from industrial espionage for instance.

One security mechanism would probably not be able to cover all the security requirements for all traffic classes, unless the strongest security mechanism is deployed. For instance, if IPsec is configured to use the AH (Authentication Header) [121] mode for all traffic classes, onboard users would likely hesitate (even refuse) to access passenger services provided by the airline (which could have an impact on the business plan of the company). Indeed, AH does not provide data confidentiality, which is actually an issue if a passenger is entering some kind of confidential data (e.g. passwords, credit card ID).

Besides, using a unique and heavy security mechanism could be expensive and low performing solution, considering the limited bandwidth for air-ground communications and the security data overhead. Consequently, as for differentiated QoS, the deployed security framework should be more flexible by taking into account security requirements that are not necessarily the same for all traffic classes.

7.3. Defense in depth

In Section 6, it has been shown that each security layer has its own advantages and shortcomings. In most cases, these advantages and shortcomings are complementary when these security layers are compared (e.g. TLS vs IPsec PEP compatibility). Then, an alternative solution is to use a defense in depth security to manage data link communications. The idea behind defense in depth is to use multiple security strategies at several layers, so that if one security layer falls down, there will be always a security backup layer to prevent a full breach into the communication system. Consequently, having a security framework that takes the advantages of each security layer and uses them to deal with the drawbacks of other security layers will be perfect. Nevertheless, such in depth security framework should not be used roughly, and must provide enough accuracy to choose the security layer mechanism only when needed. Defense in depth security has been pointed out as an important approach in order to mitigate security risks. Karmarkar and Martin [112] provided a defense in depth layered approach in order to secure the communication network against attacks. Stephens [122] identified SWIM messaging services security as an important additional block to achieve defense in depth.

7.4. ATM scalability

As discussed in the introduction, the growing air traffic and passenger load is now well established. In order to manage all the security primitives that might be needed by aircraft, passengers, or avionic systems, a PKI has to be deployed. PKI is usually defined as a set of practices, technologies, and policies involved in several processes such as deployment, management, storage, and revocation of certificates when cryptography is used. The aim is to create a “chain of trust” between heterogeneous entities in order to provide basic security services such as authentication. In the aeronautical environment, the management of PKI credentials gets more complicated because of the typical constricted network capacity of air-ground technologies: both signaling and data messages induced by the PKI have to be performed at lower cost. For instance, retrieving large CRLs repeatedly can be a bottleneck if the aircraft does not use caching mechanisms. Another restricting factor, but not the least, is the aircraft mobility. Indeed, since the aircraft and passengers should get seamless services during the flight, onboard entities should be able to authenticate or exchange required certificates when needed.

Because different aviation organizations may have different security policies in their own PKIs, interworking and roaming schemes between aircraft, end entities, or airlines are required. In such conditions, deploying a PKI regardless of these considerations becomes a tough task. Thus, a great challenge lies in finding adequate mechanisms and a well-suited PKI for data link communications under such constraints.

7.5. Impact of security on air-ground resources

Security has an undeniable impact on the network performances. Experimental studies have been carried out in the past and demonstrated the performance impact due to security protocols such as IPsec [123] or SSL (Secure Socket Layer) [124]. Generally, security should be optimized, otherwise it may induce heavy overhead to data transmission and could deteriorate the system performances. For instance, public-key authentication may have an impact on the end-to-end delay [125] whereas encryption/decryption schemes consume computational and network resources. In terrestrial networks such as LANs (Local Area Networks), large bandwidths are usually available. This makes easier the deployment of security mechanisms, regardless of the induced overhead. In contrast, air-ground data link systems provide limited capacities. On top of being limited, these bandwidth channels are shared among several traffic classes as a consequence of service integration on the same satellite link and operational services RCTPs (Required Communication Technical Performances)⁶ are very strict [9]. Therefore, specific techniques shall be used to optimize the security solution

⁶ RCTPs have been established by ICAO and FAA to define technical performances that every aeronautical communication system must achieve. These RCTPs can be expressed for instance in terms of end-to-end delay, latency, service priority, or security properties.

for the mobile bandwidth-limited aeronautical communication environment.

Meanwhile, QoS information should play a key role in order to minimize the security overhead. Indeed, information about available network and computational resources are precious in view of choosing the security algorithm that fits the best to the current context. For instance, knowing how much bandwidth capacity is left for a given traffic class could be an interesting information in order to secure the communication. Thereby, the security could be adjusted to the network state and available resources while satisfying security requirements for each traffic class. In such a case, designing an adaptive security framework should be a valuable solution for the aeronautical context. Finding the best trade-off between required security services and system performances may be an interesting challenge to meet. From a conceptual point of view, this is equivalent to a knapsack problem: the aim is to maximize security while the induced data overhead is minimized as much as possible.

7.6. Vulnerability and risk assessment

Risk assessment is generally considered as the core of the computational framework in a risk management process for a network information system. This process is mandatory and crucial for the protection of interconnected systems that provide various services to their clients or users. Traditionally, system vulnerabilities are identified, determining the likelihood of occurrence of threats being exploited, and evaluating the consequences of attacks that could take advantages of these security holes. Having assessed the risks, security measures (which could be technical or operational) are identified then implemented in order to mitigate those risks.

With the rapid introduction of data link systems, a risk assessment process should be conducted when the security of the network is analyzed. For the specific aeronautical context, there is no mandatory methodology imposed by aviation standard organizations, withal some risk assessment studies have been carried out and followed nearly the same principles. For instance, the ARINC 811 report provided a security process framework to meet airline specific needs. ARINC 811 is mainly dedicated to AOC services.

Another example of a risk assessment analysis in the field of aeronautical communications has been conducted throughout the NEWSKY project. Indeed, the threat analysis approach used in the project relied on a domain-based security methodology, known as DBSy (Domain Based Security). This approach is based on data exchanges between different security domains. In the scope of the NEWSKY threat analysis, ATS, AOC, AAC, and APC communication classes have been identified as separated security domains for the study. As long as the on-board network domain is organized in several entity clusters that have different security needs and objectives, a domain-based security analysis seemed to be a sound strategy to assess the risk in an aeronautical information system architecture.

However, such risk assessment methodologies share the same shortcomings. First, the security analysis in both methodologies is qualitative. Indeed, security experts evaluated the potential likelihood and impact of each of identified

threats to determine the risk level for the network system. This is an essential issue as security expertise is expensive financially speaking, and relatively slow compared to an automated risk assessment procedure: several workshops and appointments have to be held in order to catch the subjective assessment of the experts. Finally, the measured risks are based on a ranking scale. For instance, in the case of the NEWSKY study, the likelihood of occurrence of a threat has been ranked from “*extremely improbable*”, to “*highly probable*”. Using such an evaluation methodology allows possibly to compare two different risk levels (for instance, between high and low), but it is impossible to estimate the distance between them (for instance, between two security levels ranked as high). This can be confusing for a security network administrator willing to improve the overall security level of the network.

Besides qualitative issues, these risk assessment methodologies omit that an intrusion is mostly a transitive attack. Indeed, when an attack occurs on a network node, it is highly likely that the intruder will try to attack the interconnected nodes when this is allowed by the network topology. The attacker would be able to do so if there were some system assets that could help him to break into a connected node. These assets could be applications, services (intruded on an associated port), user logins (e.g. root privilege access), or database access accounts. In the data link context, the risk resulting from interconnection between nodes and different networks has to be mitigated because of the future SWIM (System Wide Information Management) paradigm. Indeed, the ATM has already adopted the concept of SWIM as the means to improve data exchange between various applications in different domains (e.g. flight data management, weather systems, aeronautical information management). SWIM can be seen as interconnected information management infrastructure that create a “system of systems” between new and existing ATM systems. Therefore, when the network risk level has to be estimated, it should include the risk resulting from the interconnections between the node within the network (i.e. the propagated risk between two or many interconnected nodes). Section 7.10 discusses further details about SWIM security.

7.7. ADS-B security

Surveillance technologies are also evolving from radar to data-based monitoring of the air traffic control. The ADS-B (Automatic Dependent Surveillance Broadcast) technology, which is progressively being used across the worldwide skies, allows a moving aircraft to compute its 3D position and broadcasts information (e.g. ICAO aircraft ID, position, velocity, etc.) using data link. However, ADS-B is weakly protected against loss of data integrity, which is currently considered as the main challenge when the ADS-B technology is used. As messages are not signed by the sender, attackers could create false ADS-B messages with erroneous GPS positions to cause midair collisions for instance.

In order to prevent misuse of ADS-B data, cryptography could be used. Asymmetric cryptography is preferred as it does not require any precondition knowledge between aircraft and ground entities (e.g. pre-shared symmetric keys). PKIs should be privileged using strong cryptographic schemes

such as certificates, elliptic curve derived keys, or hash-based message authentication messages. As depicted earlier, PKIs should minimize overhead, specially for the protection of ADS-B-Out data because of their small payload (e.g. 272 bits for UAT—Universal Access Transceiver).

ADS-B security is currently a central topic of several work. Sampigethaya et al. [126] provided a security analysis of ADS-B and a taxonomy to classify potential attacks on the protocol. Costin and Francillon [127] demonstrated that practical and easily feasible attacks can be conducted on ADS-B devices. McCallie et al. [128] provided a comprehensive overview of ADS-B security, see [128] for more details.

7.8. Safety and security interdependencies in ATM

Historically speaking, safety and security have always been considered as two distinguished and separated concepts. Nonetheless, there are strong dependencies between them, specially in high risk and critical situations within nuclear, electric power industry or ATM environments for instance. Moreover, the convergence of safety and security requirements brings new consequences and challenges which are still to be addressed. The ATM system is particularly prone to these ambiguities because it involves many entities from multiple disciplines with different purposes (communications, navigation, and surveillance). For instance, it is interesting to analyze the side effects of security features, added to preserve data link communications from malicious attacks, on the occurrence of accidental events and highly requested availability of the operational services. As matter of example, if an onboard firewall went off because of an erroneous command or a power-supply issue (both are not malicious events), an adversary could take advantage of the situation by intruding the aircraft system more easily.

7.9. LDACS security

Under the EUROCONTROL and FAA Action Plan 17 activities, the LDACS (L-band Digital Aeronautical Communication System) [129] has been recently identified as the next best candidate to support future aeronautical requirements in continental areas (e.g. cellular communications). Two versions of the system have been identified: the FDD (Frequency Division Duplex) based LDACS-1 and the TTD (Time Division Duplex) based LDACS-2 systems. These systems are meant to progressively replace aviation specific continental data link systems such as VDL. However, both LDACS protocol stacks do not integrate security mechanisms. This may lead to several security problems for the foreseeable LDACS usage.

7.10. SWIM security

SWIM is a new paradigm in the aviation industry. It defines a wide, modular and flexible architecture completely transparent for aviation actors. It allows information sharing (e.g. meteorological information) and operational air traffic services with a reduced overall cost of operation and maintenance. SWIM also allows interconnection of ATSPs (Air Traffic Service Providers) with external networks (e.g. Internet). Because

of such requirements, SWIM security must be tackled. Indeed, providing security in a distributed network-centric⁷ environment is a significant challenge. [83] provides an overview of the FAA SWIM security architecture testbed and explores some of the methods and mechanisms used to ensure end-to-end security to the SWIM applications and users.

8. Improvement directions for aircraft communication security

In this section, three solutions are given in order to address the data link security issues and challenges mentioned in Section 7. Each of the following subsection is dedicated to a given solution.

8.1. Adaptive security

Adaptive security should provide enough flexibility and accuracy to deal with many of the security issues previously depicted. For instance, a security manager module could be installed onboard the aircraft to manage all the secure communications between the air and the ground. As one infrastructure is meant to deal with several traffic flows coming from different network domains, such a security manager module could be deployed on a proxy that intercepts the clients requests and establishes a secure communication with the ground entity. In order to avoid a single point of failure for all the security architecture, the safety of such a security equipment should be guaranteed (i.e. using redundancy techniques).

This kind of solution has the advantage to provide a differentiated security covering all the security requirements of onboard services. Moreover, it provides a security defense in depth through the use of different security mechanisms complementing each others. To make the adaption, the security manager could use the information about the used and available system resources to activate the right security algorithms and protocols. From a networking point of view, Fig. 8 illustrates an example of a possible onboard network architecture using this kind of secure proxy.

Two proxies with adaptive security capabilities are located into DMZs (Demilitarized Zones) and connected to two ATN/IPS routers. The left-side router is relevant to the APC and AOC networks and provide QoS schemes to differentiate passenger from airline data flows. The right-side router is relevant to the ATS network providing a strict separation between ATS and others types of data flows. Secure request connections are redirected to the security proxies where each request is treated individually. Supported security mechanisms are negotiated between the proxies and the ground entities, and then an adapted security policy is established. The left-side proxy secures two types of data flows (i.e. APC and AOC), so it uses both the priorities and the system state (i.e.

⁷ Network-centric refers to an interconnection of heterogeneous entities (e.g. people, devices, services) by a communication network. The aim is to share information and optimize resource management at a global scale.

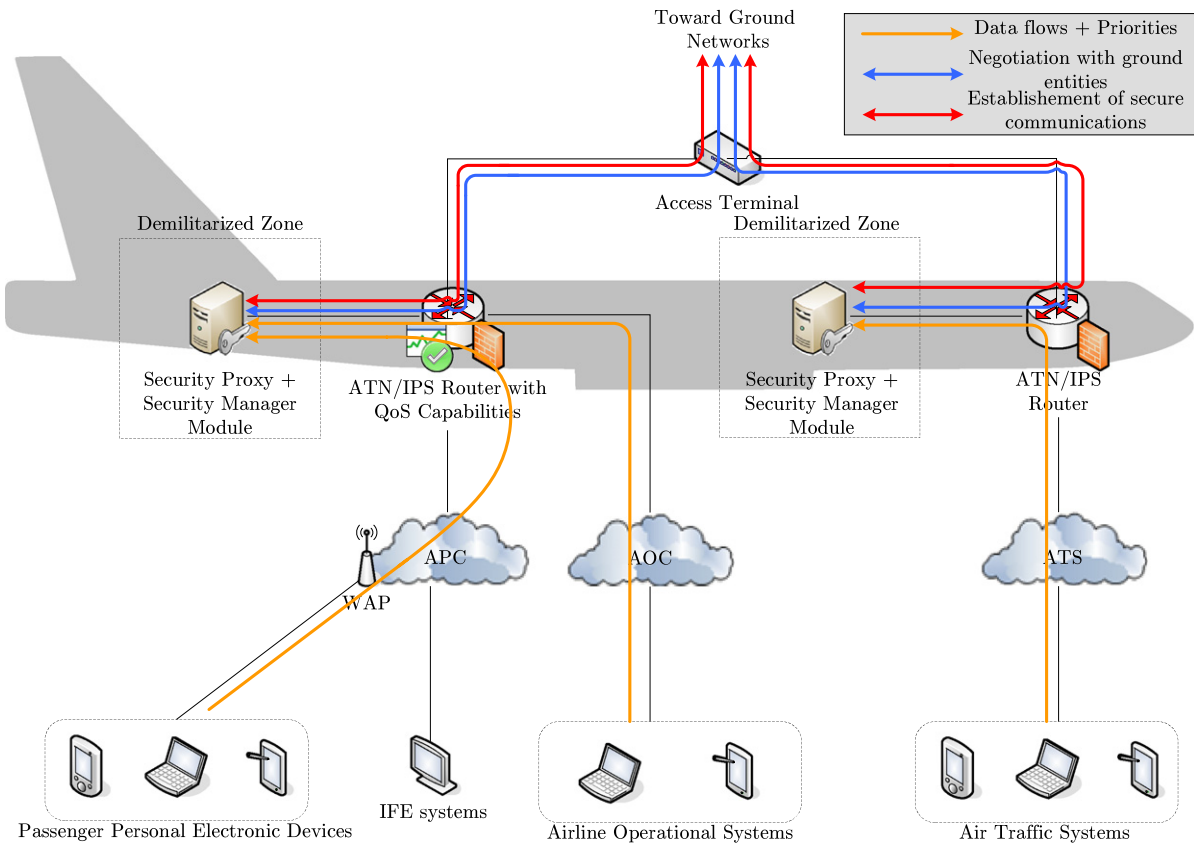


Fig. 8 – Onboard logical secure architecture using an adaptive security policy.

available network and system resources) to choose the security policies. On right-side, only ATS traffic is handled by the security proxy, thus the security policy is defined only according to the system state.

From a protocol architecture point of view, Fig. 9 illustrates a possible protocol stack for the proxy using adaptive security capabilities, meaning picking the right security mechanism(s) depending on a combination of factors such as the priority, QoS, or security requirements. The security manager module could work above the FRS (Future Radio System)⁸ boundary on the control plan. This has the advantage to provide a complete independence regarding the access layers (e.g. AFDX, AeroMACS, etc.). Thus, access layer security could be used in complement to the security policy selected at higher layers of the protocol stack. Besides, it represents a good solution to mitigate ADS-B or LDACS security issues discussed in Section 7. Besides, heterogeneity of security requirements and priorities between data flows discussed earlier are mitigated with such an adaptive security solution.

As shown in Fig. 9, the inputs for the adaptive security proxy are:

- *Data flow priorities*: every data stream is redirected to the security manager proxy with the priority associated to its

traffic class (ATS, AOC, or APC). These priorities are combined with security requirements and QoS considerations (see below) in order to activate the right security mechanisms on the proxy protocol stack;

- *Security requirements*: security requirements for operational services have been identified in the COCR (Communications Operating Concept and Requirements for Future Radio System) document [12]. For non-operational services, security requirements are usually expressed by the final users;
- *QoS considerations*: information relevant to the available system resources might be retrieved then used when a security policy has to be established by triggering security mechanisms of the TCP/IP protocol stack.

The security proxy design might also rely on a cross-layer approach in order to exchange the QoS, security, and data service information, collected at different layers of the TCP/IP protocol stack. The adaptive security manager module could be seen as a shared component by all the layers where information received from the protocol stack can be regrouped and used properly. Unlike other cross-layer design techniques (e.g. upward/downward information flow, merging of adjacent layers, etc.), using a transverse control module has the advantage of not bringing any modification to the classical layered design of the TCP/IP model [130].

⁸ The physical implementation of the radio components of a communication system are collectively referred to as FRS in the COCR [12].

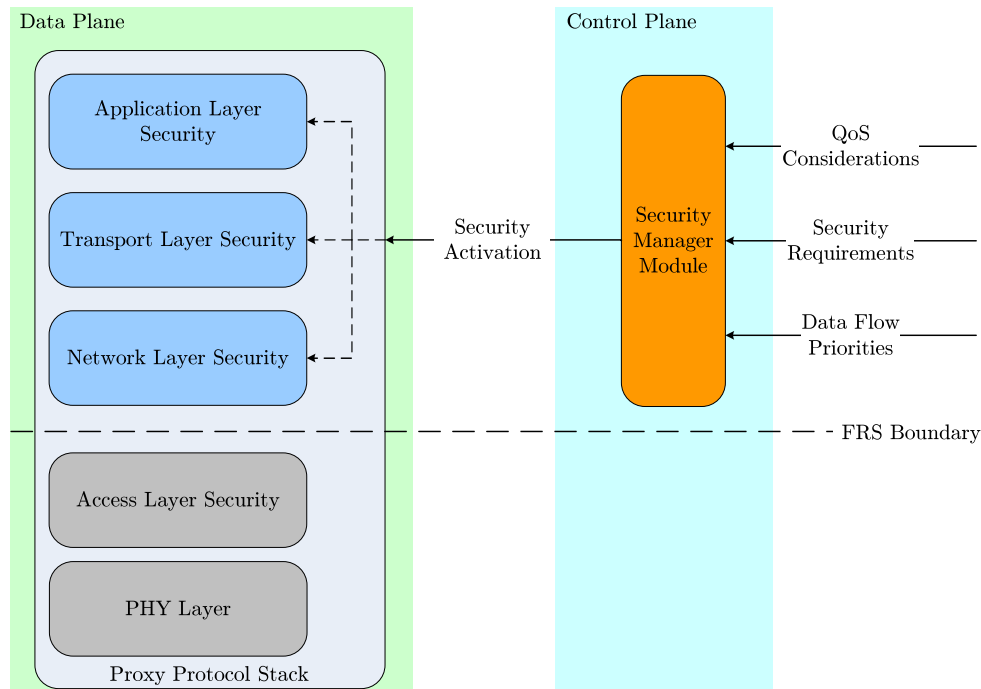


Fig. 9 – Security manager proxy protocol architecture.

8.2. Enhanced vulnerability and risk assessment

In order to estimate the security risk, the following expression is usually used [131]:

$$\text{Risk} = \text{Likelihood} * \text{Impact}$$

Where likelihood is the probability of occurrence of a threat and impact is the potential damage resulting from the exploit of the associated vulnerabilities. However, when an attack occurs on a network node, it is highly likely that the intruder will try to attack the interconnected nodes when this is allowed by the network topology. The attacker would be able to do so if there are some system assets that could help him to break into a connected node. These assets could be applications, services (intruded on the associated port), user logins (e.g. root privilege access), or database access accounts. Strong dependencies between these system facilities imply some kind of transitivity in the network risk propagation process.

By way of example, let i and j be two correlated nodes in the network and t an exploitable vulnerability on node j as shown in Fig. 10. Since node j has some vulnerabilities that could be exploited by an attack (step B), it might transmit its correlative risk to the connected node i (step C). This risk will propagate to the different nodes connected with node j . Besides, as long as the risk has been propagated from node j to the correlated node i , there is a strong probability that the intruder continues his way and tries to break in nodes connected with i (node k in step D).

In such a case, the classical security risk expression is no more reliable as it does not take into account such propagation properties. Then, the risk expression could be extended to integrate the propagated risk resulting from the correlation between logical nodes in the network as the following:

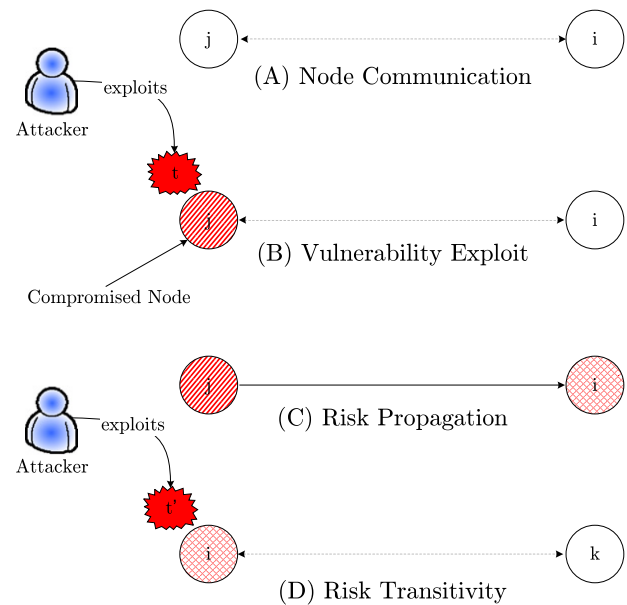


Fig. 10 – Risk transitivity between correlated nodes.

1. **The risk per node** is computed for each node depending on its own vulnerabilities and its connections with correlated nodes. As a node is connected to other nodes in the network, the total risk for a given node i is evaluated as the product of node value $Value_i$ and the sum of its individual and propagated risks (respectively denoted $Risk_i^+$ and $Risk_i^-$). The node value is computed by taking into account the node functionality which reflects the importance of a host from a security point of view, the priority of the

domain the node belongs to, and the number of connected nodes to node i . The risk for the node i is then computed as:

$$\text{Risk}_i = \text{Value}_i * (\text{Risk}_i^+ + \text{Risk}_i^-).$$

2. **The individual risk** is the intrinsic risk computed for each node, meaning it takes into account only the vulnerabilities associated with the node itself. The individual risk Risk_i^+ is computed as the sum on the number of existing vulnerabilities T_i of the product between the likelihood of occurrence of a threat $P_t(i)$ and its impact $I_t(i)$, which is fully compliant with the basic expression of the risk:

$$\text{Risk}_i^+ = \sum_{t=0}^{T_i} P_t(i) * I_t(i).$$

3. **The propagated risk** is the risk inherited from the dependency between correlated nodes (e.g. data flow exchanges, client-server architectures, etc.). The propagated risk Risk_i^+ , is estimated as the following:

$$\text{Risk}_i^+ = \sum_{j=0}^{n_i} \sum_{t=0}^{T_j} P_t(i, j) * I_t(i, j).$$

Compared to the individual risk expression, the idea is quite the same except the difference that the propagated likelihood $P_t(i, j)$ and the propagated impact $I_t(i, j)$ are induced by all the vulnerable nodes connected with node i (and denoted n_i).

4. **The network risk** is the total risk computed for all the nodes composing the network. It is calculated as the sum of all the risks relevant to each node in the network (where n denotes the total number of hosts on the network):

$$\text{Risk}_{\text{net}} = \sum_{i=0}^n \text{Risk}_i.$$

The impact of threats could use the CVSS (Common Vulnerability Severity Score) provided by the NVD (National Vulnerability Database) as it represents a quantitative manner to estimate the severity of a given vulnerability. Likelihood of occurrence of threats could use behavioral properties related to an attack (e.g. the motivation to attack a node vs. the technical difficulties to exploit the vulnerability).

8.3. Scalable public key infrastructure and key management

A performance-aware PKI should provide an efficient and scalable key management for the future E-enabled aircraft. This ATM dedicated PKI must provide three fundamental properties:

1. **Scalability** is probably the most important criteria as it should help in decreasing the amount of security overhead. When a PKI is used, several security-related procedures take place aside the effective secure exchange of data flows: registration of end entities to the CA, key generation, certificate distribution/revocation/verification, etc. As these procedures require the use of several signaling messages (e.g. request for a CRL—Certificate Revocation List), they should be minimized and optimized;

2. **Interoperability** is needed, first to provide a smooth aircraft mobility (transition from an airline domain to another for instance) and a seamless service to passengers and users onboard the aircraft;
3. **Robustness** is critical as the PKI should avoid a single point of failure (e.g. single CA) and provide a chain of trust between its different components.

In order to guarantee all three properties, a multi-rooted hierarchical PKI model with cross certification between trusted CAs may be used as depicted in Fig. 11. In order to have a trusted relationship between third party authorities with end entities, and cost-effective communications in a large scale ATM system, it is suitable for CA to manage a limited number of PKI operations.

Therefore, such a PKI model could use three levels of hierarchy:

1. **Level 1** is relevant to the inter-CA communications: a ground located root CA is deployed for each airline and is responsible for all the end entities that belong to this airline. According to the ATA Spec 42 document [38], it is clearly stated that the CA should be deployed by a trusted agent like an airline. The end entity can be on the ground (e.g. an ATN router) or on the air (e.g. an aircraft). Besides, as long as every root airline CA is independent of the others and has the authority on the aircraft labeled within its own domain, cross certification can be used between the ground trusted CAs. Cross-certification generally occurs between inter-domain CAs, meaning that these CAs do not belong to the same domain. Thus, the autonomy of local ground CAs and interaction between end entities belonging to different airlines can be always provided. Basically, when a certificate validation is needed between two end entities which do not belong to the same airline at a higher level of the PKI model, the root CAs have to cross certify, meaning a given airline CA has to sign the certificate of another airline CA. Cross certification is currently handled by CertiPath in the aviation industry [132];
2. **Level 2** is relevant to the communications between the root CA of an airline and aircraft managed in the airline domain: delegated (or subordinate) CAs are deployed onboard each aircraft and used to handle the onboard certificate entities. The basic idea is that each subordinate CA onboard each aircraft operates as a master and trusted certificate authority for all the end entities embedded in the same aircraft. In MANETs, the idea of trusted community is of common use, specially where end entities work in several clustered groups. This idea could be used as a starting point to reduce the air-ground overhead induced by PKI operations. In this model, a community is an aircraft encompassing all the end entities onboard (devices, physical entities, software, etc.) and a sub-CA (the locally trusted certificate authority);
3. **Level 3** is the last level of the hierarchy and deals with communications between end entities and sub-CAs onboard each aircraft: the sub-CA works as a trusted third party entity for the community onboard the aircraft, and is responsible of managing all the certificates of these entities. Therefore, end entities do not need to contact the ground root CAs when a certificate or a key is needed, the only

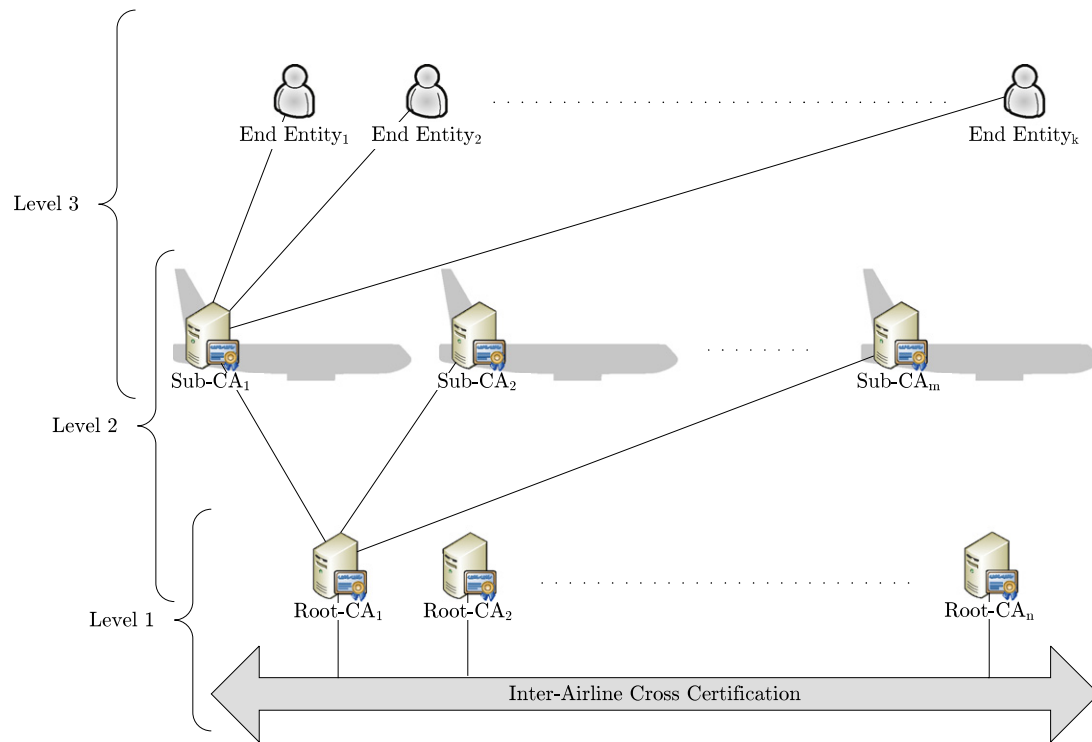


Fig. 11 – Cross-certified multi-rooted hierarchical PKI model.

Table 4 – Mapping between challenges and improvement directions in data link security.

Data link security challenges	Adaptive security	Enhanced vulnerability and risk assessment	Scalable PKI and key management
Data flow logical separation	✓		
Heterogeneity of security requirements	✓		
Defense in depth	✓	✓	✓
ATM scalability			✓
Impact of security on air-ground resources	✓		✓
Vulnerability and risk assessment		✓	
ADS-B security	✓	✓	✓
Safety and security interdependencies		✓	
LDACS security	✓	✓	✓
SWIM security	✓	✓	✓

trusted authority for them is the sub-CA onboard the aircraft. With such a PKI hierarchy, the signaling messages relevant to security operations are restricted between the airline ground CAs and the end entities.

Table 4 shows a mapping between the improvement security solutions provided in this paper and the data link security challenges identified earlier in Section 7. As stated in Section 8.1, adaptive security addresses data flow priorities and logical separation, heterogeneity of security requirements and defense in depth by providing several security mechanisms at different layers of the protocol stack, and the trade-off between security and limited network resources. Moreover, as the security manager is independent of medium access and physical layers, it is fully compatible with all the aeronautical data link systems such as ADS-B, LDACS, or SWIM. The enhanced vulnerability and risk assessment provided in Section 8.2 could be used for

any network-based data link systems, this includes, ADS-B, LDACS, and SWIM. It will be also interesting to map security and safety risk assessment techniques in order to investigate more in details the interdependencies between them. Besides, this risk assessment methodology can be seen as an additional procedure to provide defense in depth and strengthen the overall security. Finally, the aim of the scalable PKI described in Section 8.3 is not only to address ATM scalability issues, but also reinforce the security of all the data link systems of the FCI. It is also a performance-aware PKI as it has been designed to reduce the impact of key exchanges on the network resources.

9. Conclusion

Since many years, the aviation industry is evolving in every aspect. Data communications will be soon widely used and

network security must be addressed to avoid unwanted side effects. This paper is a survey of network security in the future aircraft data communications. It explains why network security should be a central point of interest in the future aircraft data communication systems and provides an overview of the efforts undertaken in order to accommodate a safe and secure air traffic environment. A taxonomy for network security potential threats is provided, then already existing security solutions are depicted according to a layered classification scheme. Finally, main security challenges are discussed then likely improvement directions are presented. These enhancement directions could be used as a starting point to provide a secure environment for data link communications.

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